

rainier3d: a reproducible digital model of the subsurface and surface of Mount Rainier, Washington

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Abstract. rainier3d is an open digital model of Mount Rainier and the West Rainier Seismic Zone (WRSZ). It covers a 70×75 km box, from the ground surface to 20 km below sea level. Every input is fetched by a script from its original archive, cached, and recorded with a checksum. Every parameter names its source in a registry, and the whole database can be rebuilt with one environment and a sequence of numbered scripts.

5 The subsurface model holds P- and S-wave speed, density and attenuation on three nested grids. Mapped geology is extended to depth by explicit rules. Near-surface hydrothermal alteration is mapped from a helicopter electromagnetic survey. A pressure-dependent rock-physics law converts the units to seismic properties, and the result is merged with the USGS Cascadia velocity model v1.7 and CRESCENT Gen0 in the wavenumber domain.

10 The model is calibrated on 1823 P and 1280 S analyst picks from 88 Pacific Northwest Seismic Network (PNSN) earthquakes. Each event is relocated in 3D in every trial model, with topography honoured. Three rock-physics multipliers and a depth-dependent correction of the regional models are fitted by Gauss–Newton iterations on P, S and S–P residuals. On held-out events the root-mean-square (RMS) residual falls from 0.121 to 0.093 s for P and from 0.317 to 0.188 s for S. An independent local-earthquake tomography confirms the Vs correction beneath Rainier.

The same grids carry the following surface layers:

- 15
- elevation, geology and glacier thickness;
 - soil thickness, two water-table estimates and streams;
 - land cover, canopy height, leaf area and biomass;
 - satellite imagery.

The model also includes GNSS strain rates and the stress that the weight of the edifice exerts at depth.

20 The derived products are distributed under CC-BY 4.0 with a command-line and Python client. The client writes them in the formats used by eikonal solvers, NonLinLoc, SPECFEM3D and the EarthScope Earth Model Collaboration.

1 Introduction

Mount Rainier is the most hazardous volcano in the Cascade Range, for two reasons. Its hydrothermally weakened edifice is covered by glaciers, and its lahars reach populated valleys (Hoblitt et al., 1998; Scott et al., 1995). Seismic monitoring

25 at Rainier relies on one-dimensional velocity models, with station corrections absorbing the effects of topography and near-surface structure. The regional three-dimensional models of Cascadia resolve the crust at wavelengths of several kilometres or more (Wirth et al., 2025; He et al., 2026). They do not resolve the volcanic edifice, its altered core, its glaciers or the shallow units exposed around it.

rainier3d connects these scales in one model with three properties:

- 30
- it is tied to the mapped geology at the surface;
 - it is consistent with regional tomography at depth;
 - it reproduces the travel times of the Pacific Northwest Seismic Network (PNSN).

The same grids carry the surface layers that describe soil, water, vegetation and ice, the geodetic strain field, and the stress of the edifice load. The model is a digital model rather than a digital twin: it assimilates no time-dependent data. Only the GNSS product is refreshed on a schedule.

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This paper describes, in order:

Table 1: Outline of the paper.

Section	Content
Sect. 2	Domain and workflow
Sect. 3	How the data are compiled deterministically, and how to rebuild them
Sect. 4	Surface layers
Sect. 5	Subsurface model, with its calibration and validation
Sect. 6	Strain and stress
Sect. 7	Geohydrology
Sect. 8	Catalogue of mass movements
Sect. 9	Access to the products
Sect. 10	Limitations

2 Domain, grids and workflow

The model uses Universal Transverse Mercator zone 10N (EPSG:32610) coordinates, with elevation in metres above NAVD88 (EPSG:5703), positive up.

- 40
- **Model box.** Easting 553–623 km and northing 5149–5224 km: the geographic box 122.30–121.40° W, 46.50–47.15° N, projected and rounded outward to whole kilometres.
 - **Summit.** Columbia Crest (4392 m) lies at easting 594.5 km, northing 5189.5 km.

- **Single source of the geometry.** All grids are derived from one configuration file (`configs/domain.yaml`), and the code never hard-codes bounds, spacing or coordinate system.

45 Properties are stored on three stacked regular grids whose spacing coarsens with depth (Table 2). Cells above the ground carry no properties. The depth of each cell below the local ground surface is stored with it.

Table 2: Model grids. The master product is one xarray DataTree in Zarr v3 format (`model.zarr`) with one node per level and a surface node.

Level	Elevation range (m)	Horizontal spacing		Cells ($z \times y \times x$)
		(m)	Vertical spacing (m)	
Surface grid	–	100	–	750×700
L1	4400 to 0	250	50	$88 \times 300 \times 280$
L2	0 to –6000	500	250	$24 \times 150 \times 140$
L3	–6000 to –20000	1000	1000	$14 \times 75 \times 70$

The workflow is a sequence of numbered scripts, S0 to S23, each run as a task of one locked software environment (pixi). Fig. 1 shows how they connect:

- the model chain (S1–S5);
- 50 – the calibration loop, in which travel-time residuals update the rock-physics parameters and the regional correction (S13, S14);
- the checks and products that read the finished model.

The domain and its sensors are mapped in Fig. 2.

3 A deterministic data compilation

55 3.1 Principles

The database is compiled by code, not by hand, following five rules.

1. **One registry.** `configs/sources.yaml` lists every data set and publication the model uses (76 entries). Each entry gives its DOI or service address, licence, the date and method of verification, and its role in the model. Every number in the configuration files names a registry key. A value chosen by the authors carries the key `m1_placeholder`, so the
- 60 unsourced values can be listed and replaced. The bibliography of this paper is generated from the same registry (`pixi run bib`), with BibTeX keys equal to the registry keys.

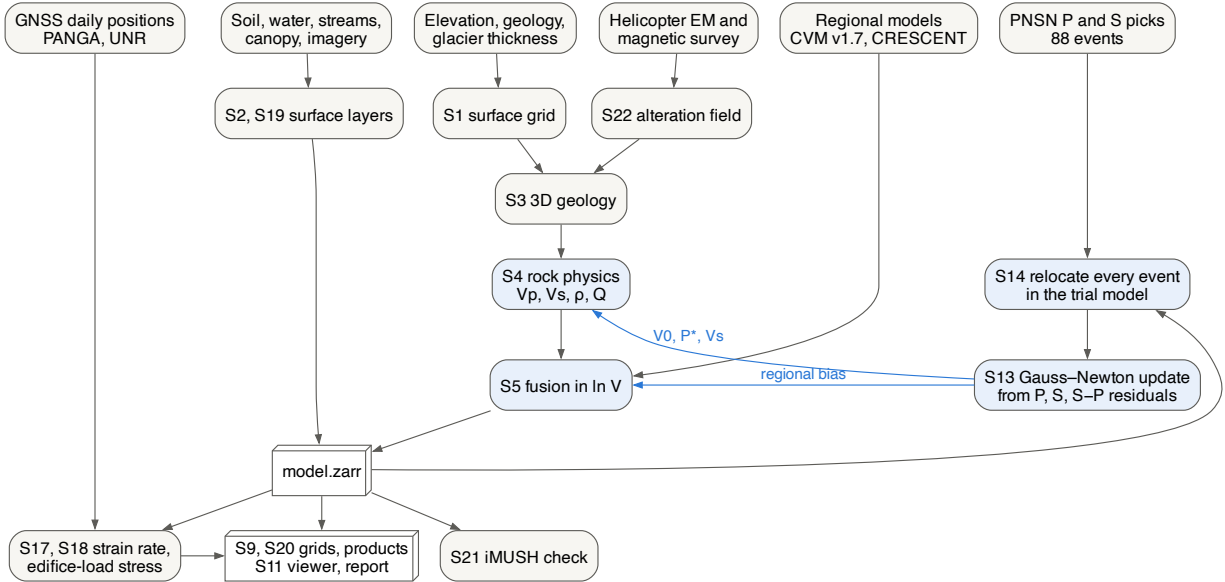


Figure 1. Workflow. Inputs (left) are fetched from their original archives. The model chain (middle) builds the surface grid, the 3D geology, the rock physics and the fusion with the regional models. The calibration (blue) relocates every PNSN event in each trial model and updates the rock-physics multipliers (S4) and a static correction of the regional models (S5) from the P, S and S–P residuals. Checks and products read the finished model (right).

2. **Original archives, cached.** Each stage downloads what it needs from the original archive, clipped to the model box where the service allows it. Examples are a window of a cloud-optimised GeoTIFF, an OPeNDAP subset, a feature-service query or a staged file. Each download is cached under `data/raw/<source>/`, and reruns read the cache.
3. **A checksum for every cached file.** `scripts/00_data_manifest.py` records the size and SHA-256 of all cached files in the committed table `docs/data_manifest.csv` (419 files, 1.48 GB; Table 3). `pixi run manifest -- --check` compares a rebuilt cache with it. Some services can return different bytes on a later request: ComCat can revise picks, and the USGS can re-stage NHDPlus. The check lists those files, so any difference between a rebuilt model and the published one can be traced to its input.
4. **One environment.** The software is pinned in `pixi.lock` for Linux and macOS (arm64). Two slim environments serve automation: `gnss` for the weekly strain refresh and `paper` for this report.
5. **Invariants.** The test suite checks the built model against rules that must hold whatever the data:
 - no properties above the ground;

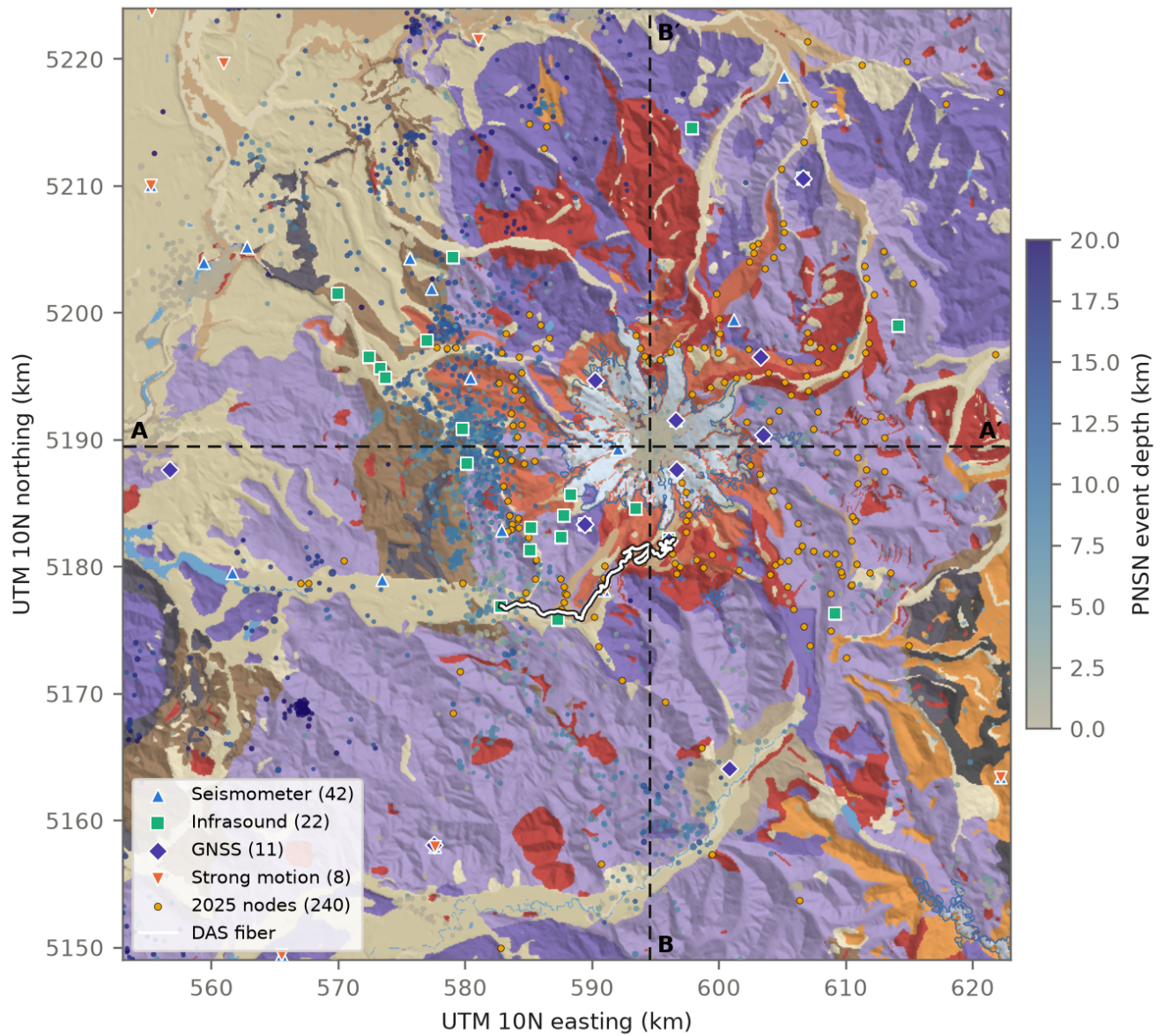


Figure 2. Model domain. The background shows the surface model units derived from the Washington 1:100,000 geologic map over a hillshade, with glacier outlines in blue. Symbols are operating sensors (FDSN metadata, GNSS site metadata and a 2025 nodal deployment). The white line is the Paradise–Nisqually Entrance distributed acoustic sensing (DAS) fibre, and dots are PNSN earthquakes since 2015 ($M \geq 0.5$), coloured by depth. The dashed lines mark sections A–A' and B–B'.

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- Vp/Vs and density within physical bounds;
 - the long wavelengths of the fused model equal those of the regional model within a tolerance;
 - the surface unit matches the unit of the top model cell.

Tolerances are fixed in the configuration and are not relaxed to make a run pass.

Table 3: The raw input cache, from `docs/data_manifest.csv`. SOLUS100 soil thickness is read directly from its cloud-optimised GeoTIFFs and is not cached.

Source folder	Stage	Content	Files	Size (MB)
dem	S1	USGS 3DEP 1 arc-second DEM, clipped	1	59
geology	S1	Washington 1:100,000 geology (GeMS), queried from the feature service	1	5
glaciers	S1	IceBoost v2 per-glacier thickness, RGI 6.0 outlines, GlaThiDa	222	44
ecology	S2	ETH canopy height 2020, NLCD 2021, clipped	2	27
hydrology	S2	NHDPlus HR geodatabases (HU4 1703, 1708, 1711), two water-table grids	7	924
imagery	S2	Sentinel-2 L2A median composite	1	96
regional	S5	Cascadia velocity model v1.7, clipped	1	60
pnsn	S6, S13	ComCat origins and phase picks (QuakeML)	93	21
sensors	S8	FDSN station and channel metadata, GNSS sites	2	1
gnss	S17	PANGA and UNR daily positions, velocity fields, manifest	80	218
emc	S21	iMUSH tomography (EMC netCDF)	1	5
rainier_aerogeophysics	S22	1996 helicopter EM and magnetic grids	8	19

Four inputs are not fetched by script:

- 80
- **Cascadia velocity model v1.7.** ScienceBase blocks scripted downloads with a captcha, so the files are downloaded by hand once.
 - **CRESCENT Gen0.** The model file is read from a local copy of its figshare record (He, 2026).
 - **PNSN one-dimensional model.** The table is read from a local file.
 - **Canopy-storage products.** The lidar canopy and soil-map products are delivered files (Sect. 4.3).

85 The code names each of these paths, and their registry entries say how to obtain them.

3.2 Rebuilding the database and the model

From a clean clone, the following commands download every input, rebuild the model and check the result. Stages that need credentials or delivered files skip those layers and say so.

```
git clone https://github.com/Denolle-Lab/mt-rainier-digital-model && cd mt-rainier-digital-model
pixi install                                # the locked environment
pixi run all                               # S1-S8: surface, layers, geology, rock physics, fusion, checks,
↳ figures
pixi run s2 -- --ma                         # optional: the Ma et al. (2026) water table (~1 GB download)
pixi run s22 && pixi run s3 && pixi run s4 && pixi run s5 # alteration from the EM survey, then
↳ rebuild
pixi run s17 && pixi run s18                 # GNSS positions, velocities, strain, edifice load
pixi run s9                                # uniform grids for ray tracing and location
pixi run manifest -- --check                # compare the rebuilt cache with docs/data_manifest.csv
pixi run test                              # unit tests and the invariants of the built model
```

The calibration (S13, about 26 min on a 10-core laptop) and the relocation comparisons (S14) are rerun as described in
90 docs/joint_calibration.md.

3.3 Redistribution

The pipeline is published in full. Derived products are published unless a source forbids it.

- **Excluded products.** A variable whose provenance includes a source flagged `redistribute_derived: false` is left out of the downloadable archives.
- 95 – **Flagged sources.** Three sources carry the flag:
 - the Ma et al. (2026) water table, whose licence (CC-BY-NC-ND 4.0) forbids derivatives;
 - the lidar canopy products;
 - the soil-map image, whose licences are not documented.
- **Rebuilding the excluded layers.** Users rebuild them with their own access. Two open archives need a free login to
100 download, NASA Earthdata for GEDI and the Copernicus Data Space for the Sentinel-2 leaf area index. Their derived layers are published.

docs/data_policy.md gives the tier of every source.

4 Surface layers

4.1 Elevation, geology and glaciers

105 Three surface layers set the top of every model column.

- **Elevation.** USGS 3D Elevation Program at 30 m, block-averaged to the 100 m surface grid. It ranges from 24 to 4380 m in the box.
- **Geology.** The Washington Geological Survey 1:100,000 surface geology in the GeMS format (Washington Geological Survey, 2026a).

- All 149 map symbols present in the box are assigned to 14 surface model units by rules on the symbol, checked against each unit’s full name in the map’s Description of Map Units. Every cell receives a unit.
- The rules are in `configs/units.yaml`, and the generated crosswalk is committed for review.
- The park map of Fiske et al. (1963) is carried as a georeferenced image for display.

- **Glacier ice thickness.** IceBoost v2 per-glacier grids (Maffezzoli, 2025) for the 219 glaciers of the Randolph Glacier Inventory 6.0 (RGI Consortium, 2017) whose centroids fall in the box, 97.3 km² in all. The grids are area-averaged onto the 100 m grid, and each glacier is rescaled so that its volume matches the IceBoost total for it. This removes an 18% overestimate from counting partly glacier-covered cells as full. The total is 5.50 km³, of which 5.32 km³ is on the cone.

The glacier bed is the ground elevation minus the ice thickness. Against the 1981 radar surveys of Driedger and Kennard (1986) as archived in GlaThiDa (World Glacier Monitoring Service (WGMS), 2020), maximum thickness agrees for Carbon, Nisqually and Tahoma glaciers (Fig. 3). IceBoost is thicker on Emmons (273 vs 185 m) and Winthrop (237 vs 98 m) glaciers. Part of any difference reflects thinning since 1981 (Sisson et al., 2011).

4.2 Soil, water, land cover and imagery

Script S2 reads each source over the model box only, resamples it onto the 100 m grid and records its source (Table 4, Fig. 4). Continuous layers are averaged over each cell; categorical layers take the most common class.

Table 4: Environmental surface layers (S2); percentiles are over the cells of the box.

Layer	Source	Native resolution	Median (5th–95th percentile)
Soil thickness	SOLUS100 depth to a lithic contact (Nauman et al., 2024)	100 m	1.41 m (0.54–1.97 m)
Water-table depth	Random-forest estimate for the conterminous US (Ma et al., 2026b)	about 24 m	10.1 m (4.8–20.4 m)
Water-table depth	Global groundwater model (Fan et al., 2017)	30'' (about 1 km)	23 m (0–306 m)
Streams	NHDPlus High Resolution, 60,174 flowlines, Strahler orders 1–8, with mean annual flow (U.S. Geological Survey, 2026b)	1:24,000	–

Layer	Source	Native resolution	Median (5th–95th percentile)
Canopy height	ETH global canopy height 2020 (Lang et al., 2023)	10 m	30 m (9–45 m)
Land cover	NLCD 2021 (U.S. Geological Survey, 2023)	30 m	evergreen forest in 72.5% of cells
NDVI	Sentinel-2 L2A median composite, 1 August – 30 September 2025 (European Space Agency, 2025)	20 m	0.85 (0.26–0.92)
NDSI	same composite	20 m	–0.48 (–0.60 to –0.24)

125 Notes on individual layers:

- **SOLUS100.** It predicts soil depth only to 2.01 m, so that value is read as “at least 2 m”.
- **Sentinel-2 composite.** It uses the least-cloudy scenes (at most 25% cloud), up to six per Sentinel-2 tile, so that every part of the box is covered. The scene classification keeps vegetation, bare soil, water, snow and unclassified pixels, and the median is taken per pixel.
- 130 – **Reflectance offset.** Since processing baseline 04.00, Level-2A reflectance is stored with an offset of +1000 that the distribution services do not remove. The offset is removed before computing indices. Without the correction, median NDVI over this forest is 0.46 instead of 0.85.
- **Snow and ice.** NDSI above 0.4 covers 66 km² at the end of the 2025 melt season, against 97 km² of glacier in the Randolph inventory outlines (about 2000). The difference is consistent with debris-covered glacier tongues (Carbon, 135 Emmons, Winthrop), which read as rock in NDSI, and with glacier retreat.

4.3 Vegetation structure

Script S19 adds the vegetation products of the canopy-storage project (M. Köpfl, University of Washington) on the same grid (Table 5, Fig. 5). They describe the canopy that intercepts precipitation and loads the ground: height, cover, leaf and plant area, and aboveground biomass.

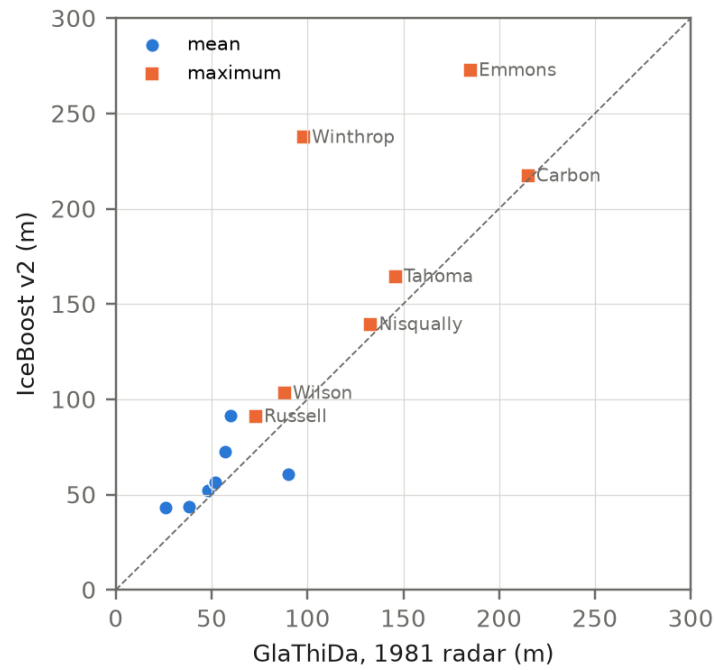


Figure 3. Mean (circles) and maximum (squares) ice thickness from IceBoost v2 against the 1981 ground-penetrating radar summaries in GlaThiDa, for seven Rainier glaciers. The dashed line is 1:1.

Table 5: Vegetation layers (S19). The provenance of the lidar and of a soil-map image delivered with these products is undocumented; both are displayed in the 3D viewer and excluded from the downloadable products.

Layer	Source	Native resolution	Median (5th–95th percentile)
Canopy height	airborne lidar, digital surface minus terrain model	10 m	14.6 m (0.3–32.4 m), on 49% of the box
Vegetation cover	same lidar	10 m	0.88 (0.01–1.00), on 49% of the box
Leaf area index	Sentinel-2 L2A, SNAP biophysical processor, 1 July – 15 August 2023 (European Space Agency, 2023)	10 m	2.36 (0.40–3.68)
Plant area index	GED1 L2B, footprints gridded (Dubayah et al., 2021b)	1 km	2.72 (0.86–3.79)

Layer	Source	Native resolution	Median (5th–95th percentile)
Canopy height	GEDI L3, mean relative height 100 (Dubayah et al., 2021a)	1 km	25.5 m (13.3–39.9 m)
Aboveground biomass	GEDI L4B, with standard error (Dubayah et al., 2025)	1 km	198 Mg ha ⁻¹ (58–436)

140 The two canopy heights differ by design. The lidar value is the mean of 10 m cells in each 100 m cell, including gaps, while the GEDI value is a 1 km mean of the tallest return per footprint. That is why the lidar median (14.6 m) is lower than the GEDI median (25.5 m) and the ETH median (30 m).

5 Subsurface model

5.1 Three-dimensional geology

145 Units are extended to depth by explicit column rules, in the manner of the San Francisco Bay region model (Aagaard and Hirakawa, 2021): the geology comes first and the velocity rules second. For a cell at elevation z and depth d below the ground, the rules of Table 6 are applied from the top down.

Table 6: Column rules for the 3D units (S3; `configs/units.yaml`).

Condition	Assigned unit
$d < 0$	air
$d < \text{ice thickness}$	ice
$d < \text{ice} + \text{deposit thickness}$	mapped surface deposit
inside the edifice footprint, above the edifice base	Rainier andesite
young volcanic rocks, or andesite outside the footprint, within 300 m of the bedrock top	that cap unit
above the base of the bedrock unit	bedrock unit (mapped, or nearest mapped basement unit)
otherwise	middle crust

- **Edifice.** The edifice footprint is the andesite and ice mapped within 12 km of the summit, closed over 500 m. Its base is the pre-volcanic surface: the elevations where basement rocks crop out around the footprint, interpolated linearly across it. The base lies at 804–2234 m over 252 km².
- **Basement.** Beneath the deposits and the edifice, the basement unit is taken from the nearest outcrop.

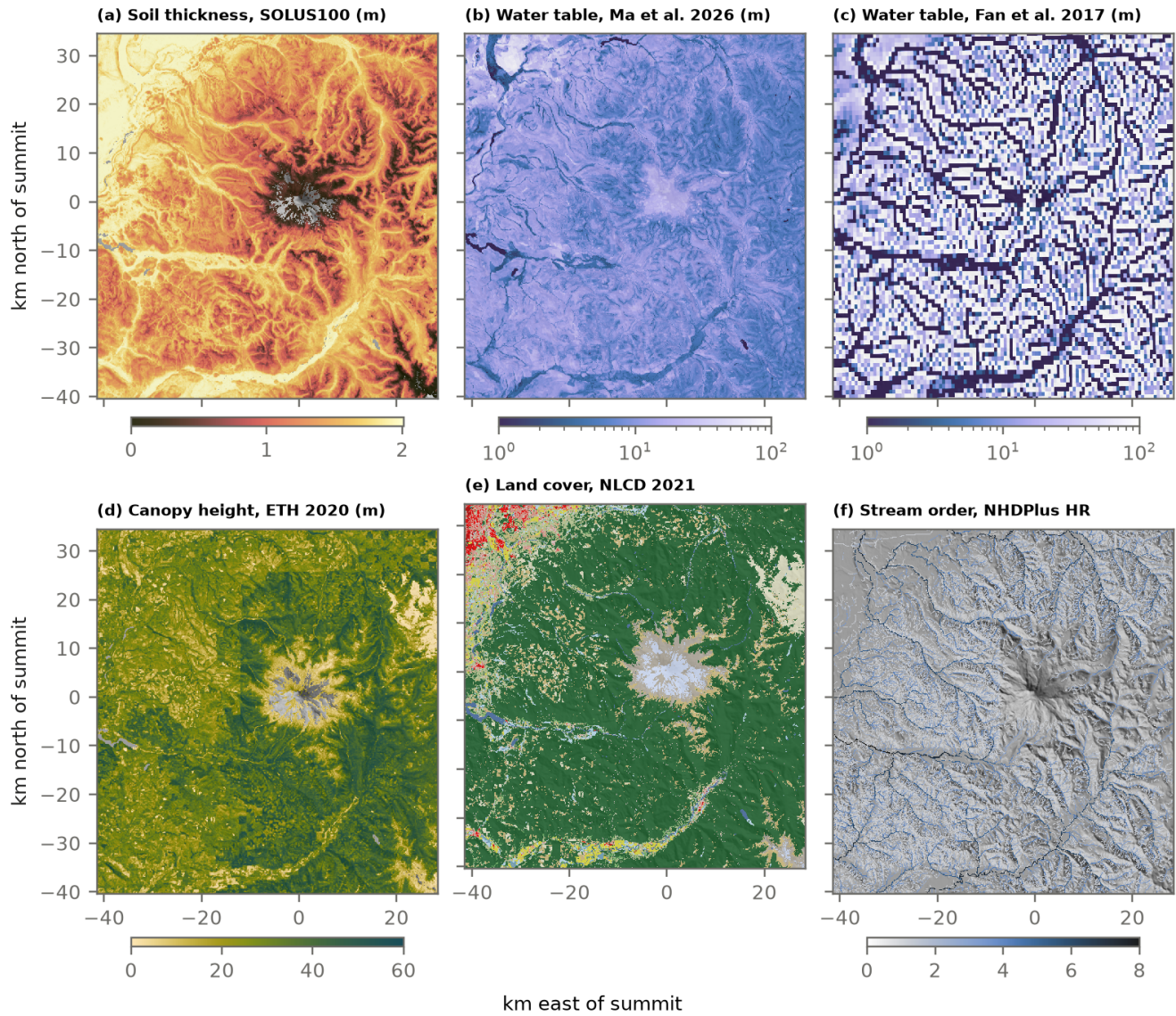


Figure 4. Environmental surface layers on the model grid, over a hillshade: (a) soil thickness, (b, c) water-table depth from two estimates on the same logarithmic scale, (d) canopy height, (e) land cover, (f) Strahler order of the NHDPlus HR flowlines. The rectangular step in (d) near 12 km west and 20 km north of the summit comes from the canopy product itself.

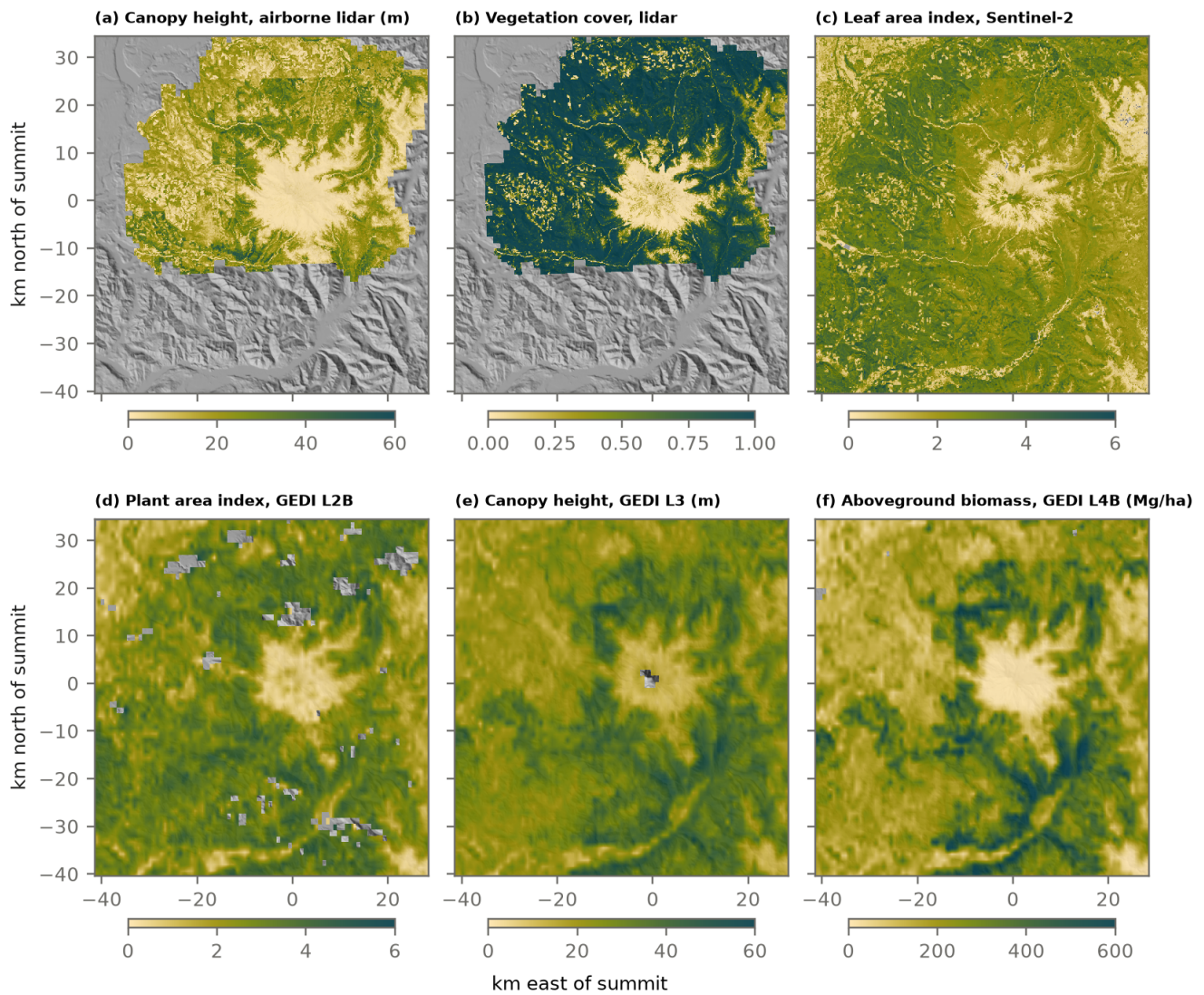


Figure 5. Vegetation layers on the model grid, over a hillshade: (a) lidar canopy height and (b) vegetation cover, (c) Sentinel-2 leaf area index, (d) GEDI plant area index, (e) GEDI canopy height, (f) GEDI aboveground biomass. Straight edges in (a) and (b) are lidar tile boundaries.

- The supracrustal units (the Ohanapecosh, Fifes Peak and Stevens Ridge formations, the Eocene volcanic rocks and the Puget Group) extend to 4 km below sea level.
- The Miocene plutons (Tatoosh, White River, Carbon River) extend to 10 km below sea level.
- The Mashel Formation is capped at 300 m thickness.

- **Unconsolidated deposits.** They have fixed thicknesses: glacial drift 15 m, alluvium and colluvium 10 m, lahar deposits 15 m.
- **Magma body.** A slow body below the summit is an ellipsoid centred 11.5 km below sea level, with semi-axes of 5 km horizontally and 6.5 km vertically. It follows the low-Vp body imaged 5–18 km beneath the summit by Moran et al. (1999).

5.2 Hydrothermal alteration from the helicopter electromagnetic survey

Clay-bearing altered rock is electrically conductive, while fresh lava and ice are resistive. The 1996 helicopter electromagnetic (EM) and magnetic survey of Rainier (Rystrom et al., 2000) measured apparent resistivity at four frequencies (33 kHz, 4737 Hz, 4341 Hz and 837 Hz) on a 50 m grid. Finn et al. (2001) used it to map collapse-prone altered zones. Script S22 turns these grids into an alteration field in three steps.

1. **Registration.** The grids are converted from NAD27 to the model datum. Their orientation is checked against the data, not assumed: the magnetic anomaly correlates with high-passed topography at 0.60 as read, and at 0.08 if flipped north–south.
2. **Intensity.** The survey saturates at a different resistivity at each frequency, so each frequency f is referred to its own fresh level L_f . L_f is the median log resistivity over the surveyed edifice lavas: 4.35, 3.89, 3.74 and 3.00 ($\log_{10} \Omega \text{ m}$). The intensity is

$$a_f = \text{clip}\left(\frac{L_f - 0.3 - \log_{10} \rho_a}{0.7}, 0, 1\right), \quad (1)$$

which starts at half the fresh resistivity and reaches 1 at a tenth of it.

3. **Depth.** Each frequency senses to a depth of investigation of half its skin depth, $503\sqrt{\rho_a/f}$ m, capped at 150 m. The three-dimensional field is the maximum over frequencies of a_f tapered below that depth. Depth is measured below the glacier bed, and the field is restricted to the edifice lavas and young volcanic rocks.

Over the 126 km² of surveyed edifice lava, 10.6 km² reaches an intensity of 0.5 or more (Fig. 6). That corresponds to 1.9 km³ of equivalent fully altered rock in the model grid. Table 7 gives the distribution around the summit.

Table 7: Alteration of the surface rock from the EM survey (S22, docs/alteration.md).

Area (edifice lavas within 6 km of the summit)	Mean intensity at the surface	Share with intensity ≥ 0.5
West flank	0.08	8%
Summit (within 1.5 km)	0.21	19%
East flank	0.27	25%

The concentration east of and around the summit agrees with the exposed east–west belt of altered rock described by Finn et al. (2001) and John et al. (2008). The EM field sees only the top ~150 m. It cannot see the altered rock buried beneath fresh cover on the upper west flank, which Finn et al. (2001) inferred from magnetic modelling. The survey’s terrain-correlated apparent magnetisation is published as a separate map.

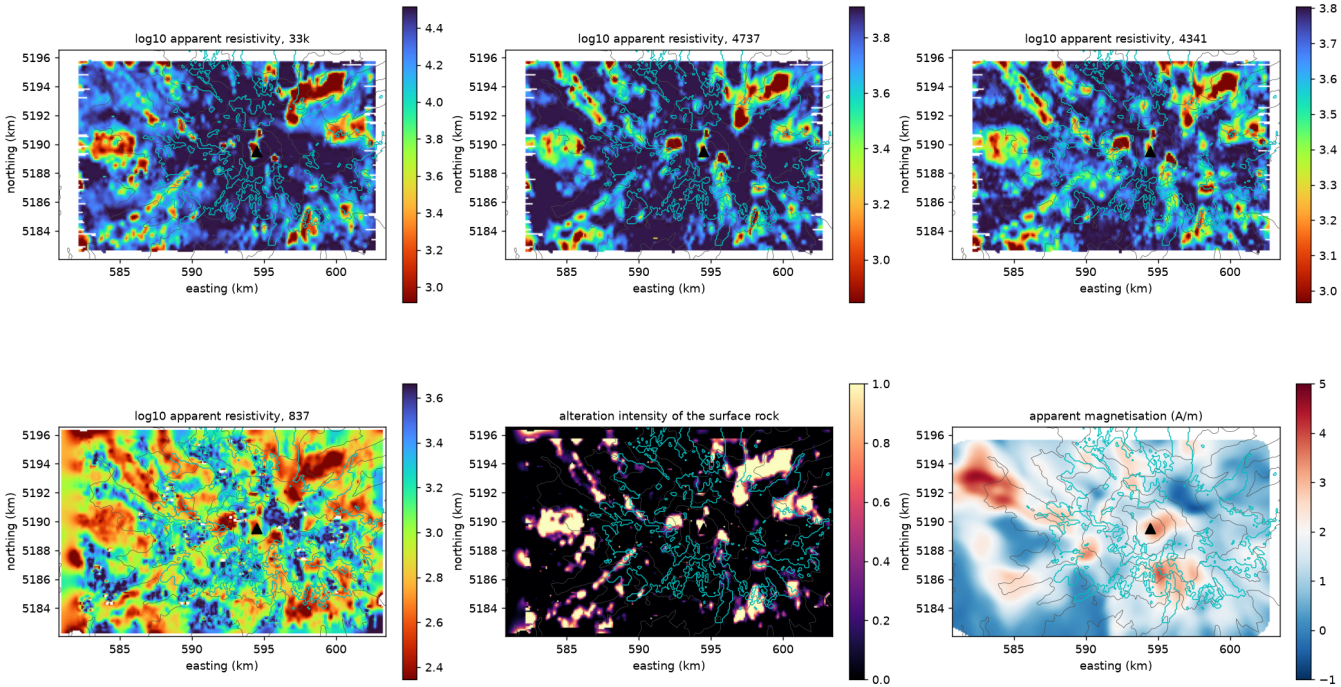


Figure 6. Apparent resistivity at the four EM frequencies, alteration intensity of the surface rock and terrain-correlated apparent magnetisation (S22). Contours: elevation every 500 m; cyan: glaciers thicker than 10 m; triangle: summit.

5.3 Rock physics

Each rock unit receives a P-wave speed that increases with effective pressure as cracks close:

$$V_P(P) = V_\infty - (V_\infty - V_0) e^{-P/P^*}, \quad P = (\rho_b - \rho_w) g d, \quad (2)$$

with bulk density $\rho_b = 2500 \text{ kg m}^{-3}$, water density $\rho_w = 1000 \text{ kg m}^{-3}$ (hydrostatic pore pressure) and d the depth below the local ground.

- **Vp/Vs and density.** Where a unit has no ratio or density of its own, both follow Brocher (2005): the regression of Vs on Vp, and the Nafe–Drake density relation (valid for $1.5 < V_p < 8.5 \text{ km s}^{-1}$).
- **Attenuation.** $Q_S = 0.05 V_S$ (Vs in m s^{-1}) and $Q_P = 2 Q_S$.
- **Parameters.** Table 8 lists the unit parameters, chosen within published ranges. The calibration of Sect. 5.5 scales them with three global multipliers. The contrasts between units therefore come from the table, and their overall level comes from the travel times.
- **Magma body.** Inside it, Vp, Vs and density are reduced by 10%, 15% and 3%.
- **Alteration.** The alteration intensity a scales Vp by $(1 - 0.30a)$, Vs by $(1 - 0.35a)$ and density by $(1 - 0.12a)$, within the ranges reported for altered volcanic rock (Heap and Violay, 2021).

Table 8: Unit parameters of the crack-closure law (`configs/petrophysics.csv`). For the rock units (Rainier andesite to middle crust, and the magma body), the calibration multiplies V_0 by 1.31 (capped at $0.98 V_\infty$), P^* by 0.91 and Vs by 0.976. Ice and the unconsolidated deposits keep their table values.

Unit	V_0 (km s^{-1})	V_∞ (km s^{-1})	P^* (MPa)	Vp/Vs	Density (g cm^{-3})
Ice	3.80	3.80	–	1.98	0.917
Glacial drift	1.60	2.40	5	2.50	1.95
Alluvium, colluvium	1.50	2.20	5	2.80	1.90
Lahar deposits	1.80	2.80	8	2.30	2.00
Rainier andesite	2.80	5.60	30	Brocher	Nafe–Drake
Young volcanic rocks	3.00	5.60	30	Brocher	Nafe–Drake
Miocene intrusive rocks	4.50	6.20	40	Brocher	Nafe–Drake
Miocene volcanic rocks	3.20	5.80	40	Brocher	Nafe–Drake

Unit	V_0 (km s ⁻¹)	V_∞ (km s ⁻¹)	P^* (MPa)	V_p/V_s	Density (g cm ⁻³)
Ohanapecosh Formation	3.40	5.90	40	Brocher	Nafe–Drake
Eocene volcanic rocks	3.50	6.00	40	Brocher	Nafe–Drake
Puget Group, Eocene sedimentary rocks	2.60	5.20	40	Brocher	Nafe–Drake
Mashel Formation	2.20	4.50	30	Brocher	Nafe–Drake
Russell Ranch Formation	4.00	6.00	40	Brocher	Nafe–Drake
Middle crust	6.00	6.50	50	Brocher	Nafe–Drake

5.4 Fusion with the regional models

Regional models. The regional model has two parts:

- **Down to 9.9 km below the ground:** the USGS Cascadia velocity model v1.7 (Wirth et al., 2025), for V_p and V_s . Its depth axis is below the ground surface. For the shallow level this is confirmed by a median top-sample V_s of 206 m s⁻¹, and for the deeper level it is inferred from continuity.
- **Below 9.9 km:** CRESCENT Gen0 V_s (He et al., 2026), with V_p from Brocher (2005).

Below 1 km, both are multiplied by a static, depth-dependent correction fitted to the PNSN arrivals (Sect. 5.5). The corrected V_s is floored at $V_p/1.6$ (Christensen, 1996).

Strategy. Neither model is right at every wavelength. The regional models resolve the crust at wavelengths of several kilometres and are constrained by teleseismic, ambient-noise and local-earthquake data; they carry no information on the mapped units. The geology model carries the mapped units and their contrasts, but its absolute level rests on laboratory-style rules. The fusion keeps each where it is informative: the regional model supplies the long wavelengths and the geology model the short ones. The combination is done in the logarithm of velocity, so contrasts are relative:

$$\ln V = \text{LP}_{\lambda_c}(\ln V_{\text{reg}}) + [\ln V_{\text{geo}} - \text{LP}_{\lambda_c}(\ln V_{\text{geo}})], \quad (3)$$

where LP is a horizontal Gaussian low-pass filter with half power at the cutoff wavelength λ_c , that is $\sigma = \sqrt{2 \ln 2} \lambda_c / 2\pi$.

Implementation details.

1. **Fused quantities.** V_s and V_p/V_s are fused rather than V_p and V_s separately, so the fused V_p/V_s always lies between the two input ratios.

- 215
- 220
- Cutoff wavelength.** λ_c grows with depth below the ground: 6 km at the surface, 10 km at 2 km, and 20 km from 10 km down. The values approximate the lateral resolution of the regional models.
 - Air cells.** Cells above the ground are excluded from the filters by normalised convolution.
 - Overshoot.** A Gaussian high-pass overshoots at sharp contrasts and puts fast rims around slow bodies. Six alternating projections remove them: each cell is clamped between its geology and regional values, then the regional low-pass is restored.
 - Top kilometre.** The top 300 m keep the geology model unchanged, and a linear taper reaches the fused model at 1 km. The regional models do not resolve this depth range; their shallowest layer is a generic near-surface model.
 - Density.** Density follows Vp through the ratio of Nafe–Drake densities, which preserves the density contrasts between units.

225

Fig. 7 shows the three models along a section through the summit. The fusion is checked by an invariant (Table 9): below 1 km depth, the low-passed fused model must equal the low-passed regional model within an RMS of 0.03 in $\ln V$.

Table 9: Departure of the calibrated fused model from the corrected regional model at the regional wavelengths below 1 km depth, and mean offset over all cells (`outputs/fusion_report.csv`). The tolerance is 0.03.

Level	RMS of LP($\ln V$) –		Mean $\ln(V / V_{\text{reg}})$,	
	LP($\ln V_{\text{reg}}$), Vp	Same, Vs	Vp	Same, Vs
L1	0.011	0.012	+0.046	+0.077
L2	0.002	0.002	+0.001	+0.001
L3	0.0006	0.0006	0.0002	0.0001

5.5 Calibration on P and S–P travel times

230

Data. The data are the analyst picks of the 88 PNSN earthquakes with $M \geq 2$ in the box between 2015 and 2026 that have at least six picks, four of them P. They comprise 1823 P and 1280 S picks at 50 stations, with origins and picks from the USGS ComCat catalogue (U.S. Geological Survey, 2026a). The event list is committed (`configs/validation_events.csv`), so every run uses the same set. Picks are weighted by the RMS of PNSN’s own location residuals: 0.14 s for P and 0.23 s for S.

Travel times. They are computed on a 500 m grid with pykonal’s point-source solver (White et al., 2020), one solve per station and phase by reciprocity.

- 235
- Solver accuracy.** Against analytic travel times on this grid, its RMS error is 8 ms in a uniform model and 11 ms with a velocity gradient; fteikpy gives 22 and 46 ms.
 - Topography.** Cells more than one cell above the ground carry the speed of sound in air, so rays follow the rock and cannot cut across valleys. The one-cell skin keeps every station in rock.
 - Cross-check.** For station OBSR, NonLinLoc’s Grid2Time P times on the exported grids agree with the pykonal times to 12 ms on average (RMS 31 ms, 85 events).

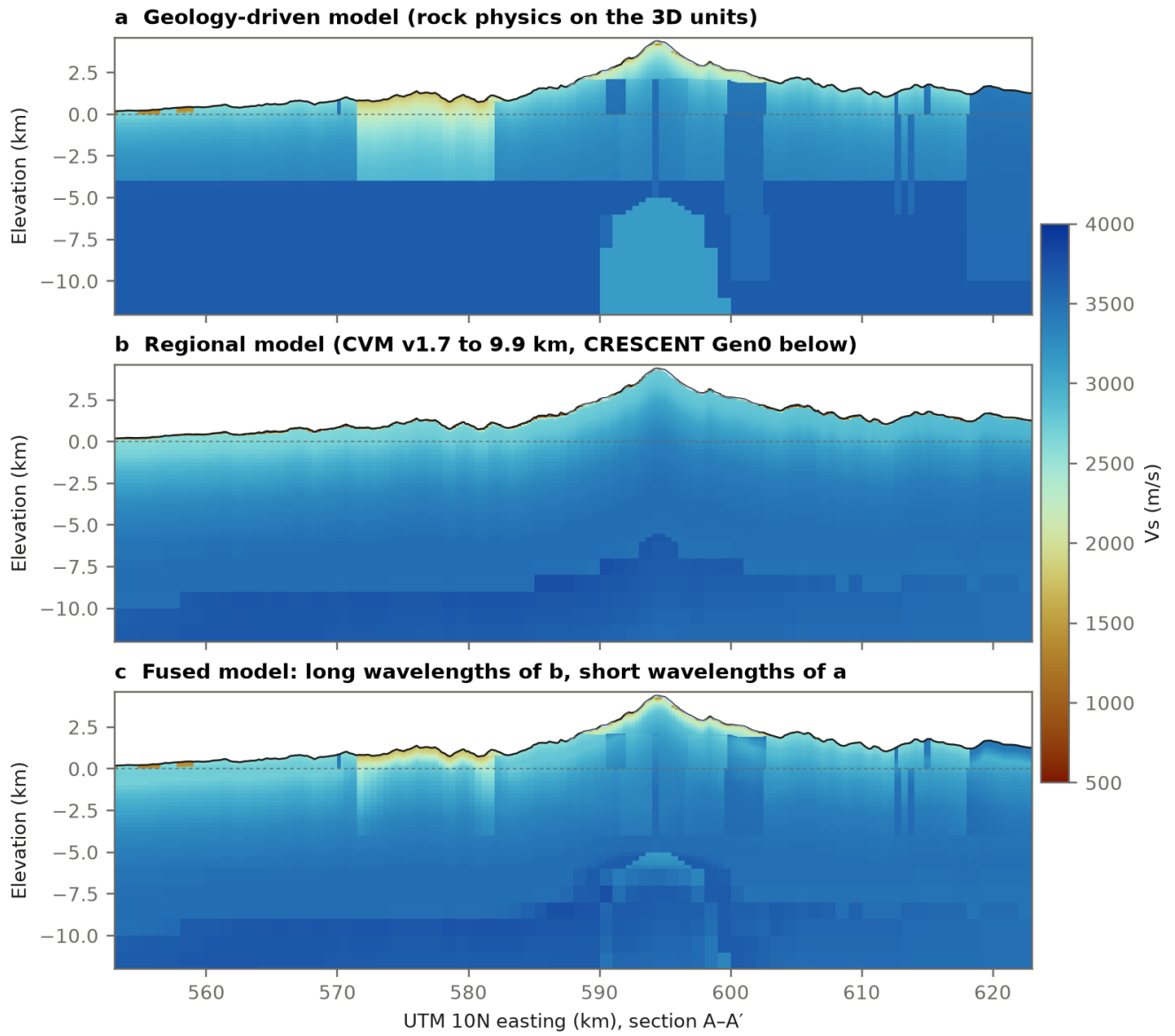


Figure 7. Vs along section A–A' (west–east through the summit): (a) geology model, (b) regional model, (c) fused model. The fused model keeps the regional long wavelengths and the geological contrasts: the Puget Group sedimentary block near 572–582 km, the Tatoosh and White River plutons, and the slow edifice.

240 **Relocation in every trial model.** A velocity model is scored with hypocentres located in that model, not with catalogue hypocentres located in a 1D model with station corrections. Each event is relocated in each trial model in two steps, in the manner of NonLinLoc (Lomax et al., 2000):

1. a grid search over the nodes at or below the ground, within 15 km of the catalogue epicentre;
2. Gauss–Newton refinement with a Huber loss (Huber, 1964) and a penalty on hypocentres more than 50 m above the ground.

245

Parameters. Fifteen parameters are fitted:

- **Geology (3).** Multipliers on V_0 , P^* and V_s of all rock units, applied in S4. Their prior standard deviations in natural log are 0.3, 0.7 and 0.1.
- **Regional correction (12).** Log-factors on the regional V_p and V_s at 2, 4, 7, 11, 16 and 25 km below the ground, applied in S5 before fusion. They are zero at 0 and 1 km, so the top kilometre belongs to the geology model.

250

Inversion. Each trial model is built by running S4 and S5 in memory. With zero calibration, this reproduces the uncalibrated model exactly.

- **Derivatives.** For the geology parameters they are finite differences through S4, S5 and the eikonal solver. For the regional correction they are ray integrals (Thurber, 1983).
- **Parameter separation.** The partials of each event’s hypocentre and origin time are projected out of the velocity update (Pavlis and Booker, 1980), and relocation and update alternate as in the minimum one-dimensional model of Kissling et al. (1994).
- **Validation split.** Events alternate between a fitting half and a held-out half in origin-time order. The smoothing weight of the regional correction (0.1) is chosen on the held-out half. Held-out events are relocated in every iterate.
- **Iterations.** Four iterations on the fitting half are followed by two on all events (Table 10).

260

Table 10: Gauss–Newton iterations of the calibration (S13, `outputs/joint_calibration/history.json`): RMS after relocation and the geology parameters. Most of the misfit reduction happens in the first iteration.

Iteration	P, held out		S, held out		$\ln V_0$	$\ln P^*$	$\ln V_s$
	P, fitting (s)	(s)	S, fitting (s)	(s)			
0 (uncalibrated)	0.124	0.121	0.314	0.317	0	0	0
1	0.099	0.094	0.195	0.189	0.131	0.268	0.017
2	0.097	0.093	0.192	0.188	0.162	0.297	0.033
3	0.097	0.092	0.193	0.188	0.267	0.403	0.016

Iteration	P, held out		S, held out		$\ln V_0$	$\ln P^*$	$\ln V_s$
	P, fitting (s)	(s)	S, fitting (s)	(s)			
4	0.097	0.093	0.192	0.188	0.256	0.176	−0.003
final (all events)	0.095	0.092	0.190	0.188	0.270	−0.099	−0.024

The fit is resolved as follows (Table 11, Table 12, Fig. 8, Fig. 9).

- **V_0 .** It is the parameter the data require: $\times 1.31 \pm 0.02$. The table rock is too slow near the surface. With the multiplier, surface V_p is 3.7 km s^{-1} for Rainier andesite and 4.5 km s^{-1} for the Ohanapecosh Formation. Because the multiplier is global, the Miocene plutons reach the cap of $0.98 V_\infty$ and are nearly uniform from the surface down.
- **P^* .** It is not resolved: its log-multiplier ranged from $+0.40$ to -0.10 across the iterations while the misfit changed by less than 1 ms, and it trades off against the regional correction at 2 km.
- **Rock V_s .** It falls by $2.4 \pm 1.2\%$.
- **Regional correction.** It changes V_p by less than 3.5% at every depth. It makes V_s 5–9% faster at 2–7 km and 5–8% slower at 16–25 km, which lowers V_p/V_s at 2–4 km depth from 1.83 to 1.74.
- **Agreement between the two models.** After calibration, the geology model and the corrected regional model agree to within 3.5% between 0.3 and 4 km depth. Uncalibrated, they differ by up to 24%, and the fusion invariant exceeds its tolerance in L1 (0.032).

Table 11: Geology multipliers (`configs/velocity_calibration.yaml`, `block geology`). Posterior standard deviations are linearised and scaled by the reduced χ^2 .

Parameter	Multiplier	Posterior s.d. (ln)	Resolved
V_0 , zero-pressure V_p	1.31	0.018	yes
P^* , crack-closure pressure	0.91	0.098	no
V_s of rock units	0.976	0.012	marginally

Table 12: Static correction of the regional models (block `regional_bias`). Formal standard deviations of the log-factors are 0.003–0.016. The deepest knots rest on few rays: 30 events are deeper than 11 km and 8 deeper than 16 km.

Depth below ground (km)	2	4	7	11	16	25
Factor on regional Vp	1.022	1.005	0.968	0.975	0.979	0.973
Factor on regional Vs	1.049	1.086	1.060	0.990	0.952	0.925

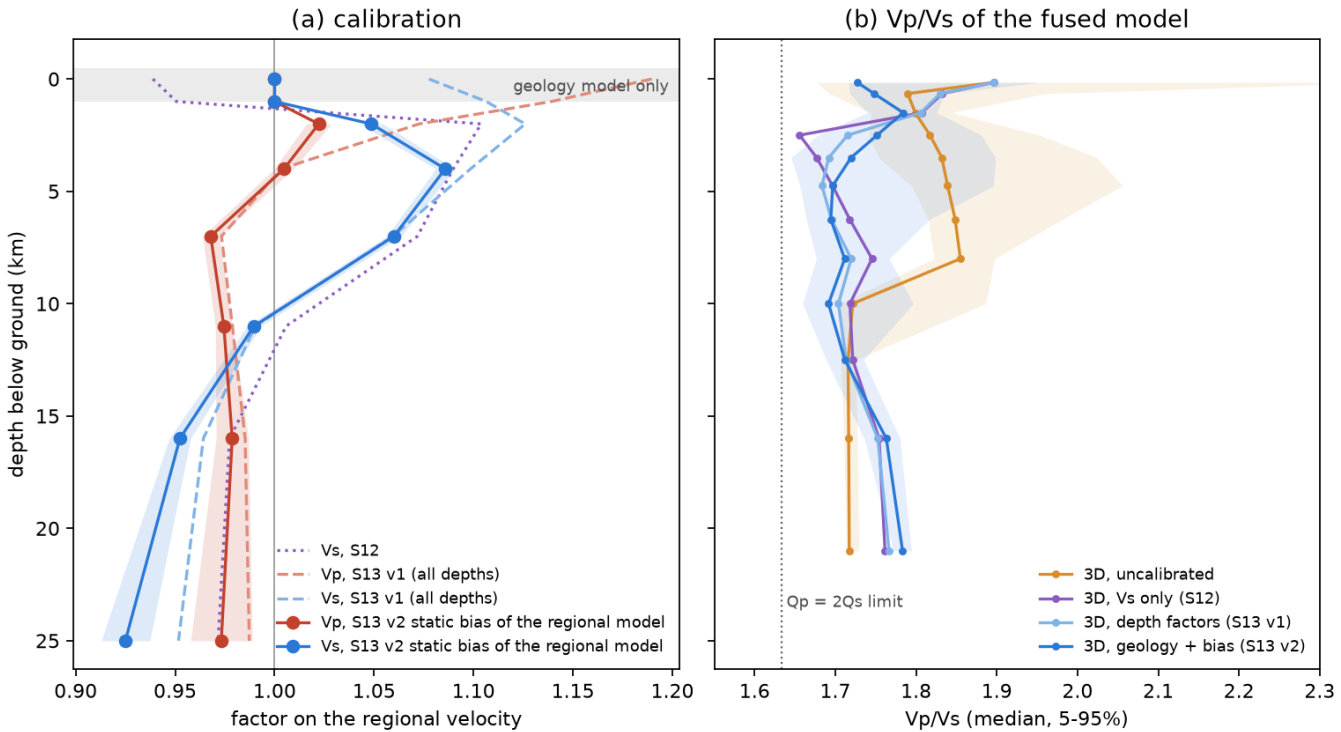


Figure 8. (a) Static correction of the regional Vp and Vs (solid, ± 1 standard deviation). The dashed and dotted lines are the two alternative parameterisations of Table 13. The grey band is the geology-only zone. (b) Median and 5–95% range of Vp/Vs of the fused model against depth below ground.

Why this parameterisation. Two simpler parameterisations were tested against the same data (Table 13).

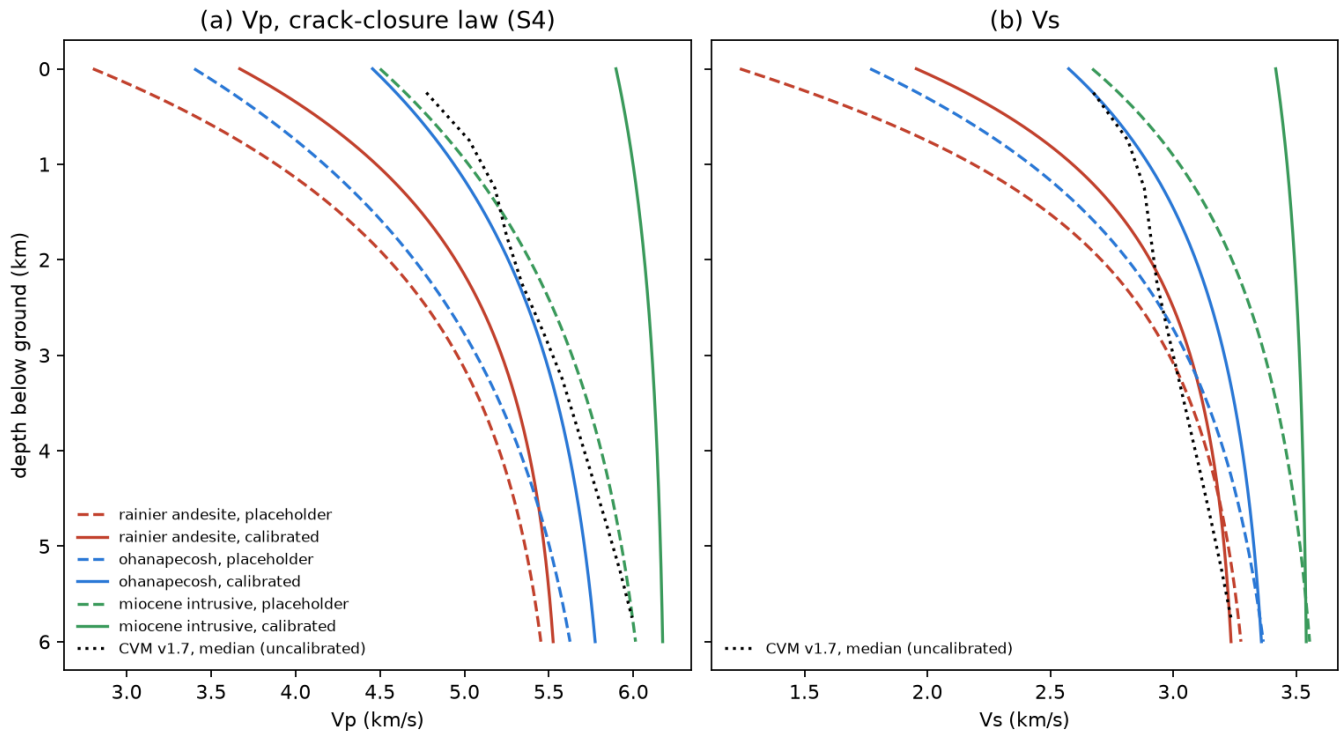


Figure 9. Crack-closure V_p (a) and V_s (b) against depth below ground for three rock units, with table (dashed) and calibrated (solid) parameters. Dotted: median of the Cascadia velocity model over the domain.

- **Vs-only depth factor, catalogue hypocentres fixed.** It fits S as well as the adopted model, but pushes V_p/V_s to 1.60–1.67 at 2–4 km. That is low for crustal rock and lower still than expected beneath a volcano with a hydrothermal system: the S–P misfit of fixed hypocentres is partly a location error.
- **Depth factors on the regional V_p and V_s at all depths, with relocation.** It fits as well as the adopted model, but places the correction in the regional model where the fusion does not use it. It requires V_p 7–19% faster in the top 2 km, where the fused model follows the geology, and it breaks the fusion invariant (0.050).
- **Adopted parameterisation.** Fitting the geology’s rock physics and correcting the regional model only below 1 km fits equally well, satisfies the invariant (0.011 for V_p , 0.012 for V_s) and keeps the unit contrasts in physical parameters.

Alteration and the calibration. The calibration was run with a conduit-centred alteration field. Replacing it with the EM-based field of Sect. 5.2 changes the relocated RMS from 0.095 to 0.093 s for P and leaves S at 0.190 s, so the calibrated parameters are kept.

Held-out events. On the 44 events not used in the fit, relocated in each model, the RMS falls from 0.121 to 0.093 s for P and from 0.317 to 0.188 s for S. The fitting half ends at 0.097 and 0.192 s, so the fit does not overfit.

All models scored the same way. Table 13 relocates all 88 events in each model with the same locator and topography. Relocation alone does not rescue the uncalibrated three-dimensional model: its S residuals stay larger than those of the 1D model. The S–P misfit therefore lies in the velocity model, not in the catalogue hypocentres. With the ground bound, no event is placed at the top of the grid, and two of the 88 events end within 50 m above the ground.

Table 13: Residuals after relocating all 88 events in each model (S14, `outputs/relocation/<model>/stats.json`). Depth shifts are relocated minus ComCat, positive deeper. ¹ Fitted on these events; the held-out scores are the fair measure.

Model	P RMS (s)	S RMS (s)	Epicentre shift	
			from ComCat, median (m)	Depth shift, median (m)
PNSN 1D	0.132	0.268	1157	+511
3D, uncalibrated	0.122	0.316	1285	+257
3D, Vs-only depth factor ¹	0.102	0.190	847	+306
3D, depth factors on regional Vp and Vs ¹	0.094	0.190	835	+727
3D, geology multipliers and regional correction (adopted) ¹	0.095	0.190	856	+852

Independent tomography. The iMUSH project imaged Vp and Vs around Mount St. Helens from local earthquakes and explosions recorded by a 70-station array (Ulberg et al., 2020; Incorporated Research Institutions for Seismology, 2020).

- **Coverage.** Its grid spans the whole model box. Its Vp is resolved over 82–90% of the box. Its Vs is resolved over 83–93% of the southern strip (south of 46.65° N) but only 11–19% of the northern half. Script S21 samples our models at its 51,300 resolved nodes below ground.
- **Beneath Rainier, north of 46.75° N (Table 14).** The iMUSH model finds the Cascadia model’s Vs 5–8% too slow at 1–8 km and 3–4% too fast at 11–20 km, the shape of Table 12. With the correction, the difference in Vs is 2% or less below 2 km. Vp/Vs moves from 1.83–1.85 towards the iMUSH value of 1.75–1.77; at 4–8 km our Vp/Vs (1.705) is about 0.05 below it.
- **South of 46.65° N, inside the iMUSH array.** The iMUSH Vs is within 1–2% of the Cascadia model at 1–6 km, and the correction makes our Vs 5–7% too fast. The correction is therefore local to Rainier.
- **Independence.** Both studies use PNSN arrivals from 2015–2016. The iMUSH-only stations and explosions make the comparison largely independent in the south, less so in the north.

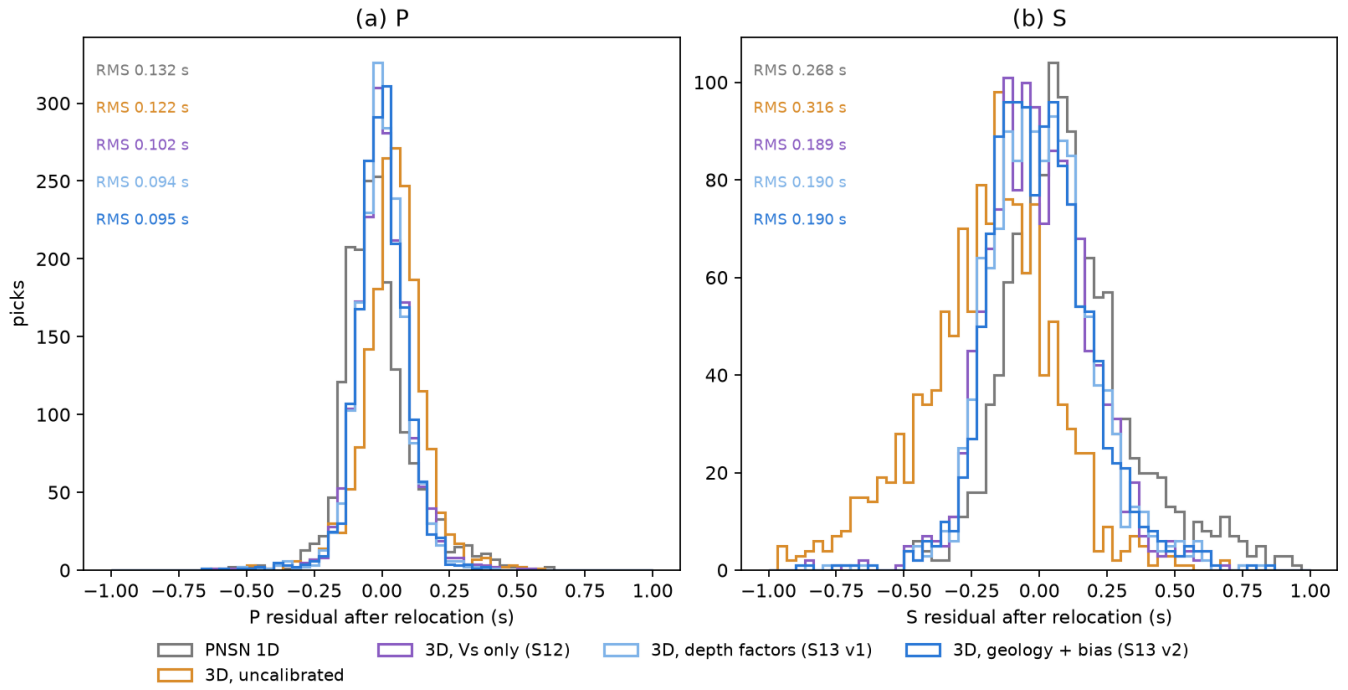


Figure 10. P (a) and S (b) residuals after relocating all events in each model; the RMS of each model is printed in its colour.

Table 14: North of 46.75° N: mean $\ln(V_{iMUSH}/V_{model})$ for Vs and median V_p/V_s (S21, outputs/model_comparison/ulberg2020_by_depth.csv).

Depth below ground (km)	1–2	2–3	3–4	4–6	6–8	11–15	15–20
Vs, regional	+0.084	+0.073	+0.064	+0.055	+0.046	−0.030	−0.035
as distributed							
Vs, regional	+0.061	+0.016	−0.010	−0.019	−0.009	−0.005	+0.017
with the correction							
V_p/V_s , iMUSH	1.770	1.759	1.754	1.757	1.755	1.744	1.765
V_p/V_s , regional	1.810	1.828	1.850	1.851	1.851	1.720	1.723

Depth below ground (km)	1–2	2–3	3–4	4–6	6–8	11–15	15–20
Vp/Vs, rainier3d	1.794	1.767	1.736	1.705	1.705	1.720	1.772

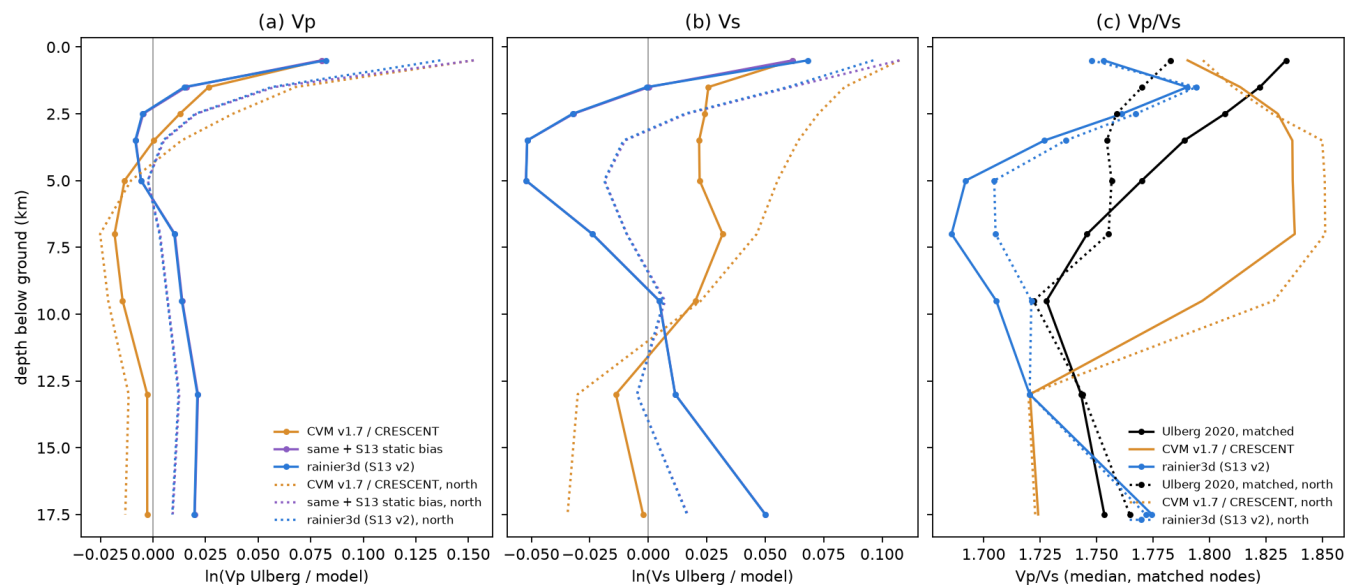


Figure 11. rainier3d and the regional models against the iMUSH tomography. (a, b) Mean $\ln(V_{\text{iMUSH}}/V_{\text{model}})$ against depth below ground, over the box (solid) and north of 46.75° N (dotted). (c) Median Vp/Vs where the iMUSH P and S models are both resolved.

305 **Catalogue hypocentres.** At the ComCat hypocentres, which were located in a 1D model, the three-dimensional model predicts later arrivals than the 1D model at every station: by 0.30 s on average for P and 0.48 s for S. With each event's mean residual removed, the RMS is 0.132 s (P) and 0.227 s (S) for the 3D model, against 0.159 and 0.287 s for the 1D model. The per-station 3D – 1D delay correlates with the mean 1D residual at 0.67 for both phases (Fig. 12). The 3D structure therefore explains part of what station corrections absorb.

310 5.7 The fused model

Figs. 13, 14 show the fused model along two sections through the summit, and Fig. 15 compares vertical profiles.

- **The top kilometre.** L1 is on average 5% faster in Vp and 8% faster in Vs than the regional model. The calibrated rock is stiffer than the shallowest layer of the Cascadia model, a generic near-surface model with a median Vs of 206 m s^{-1} . The travel times constrain this difference only beneath the stations.
- **Plutons.** The Miocene plutons stand out as fast columns to 10 km below sea level.

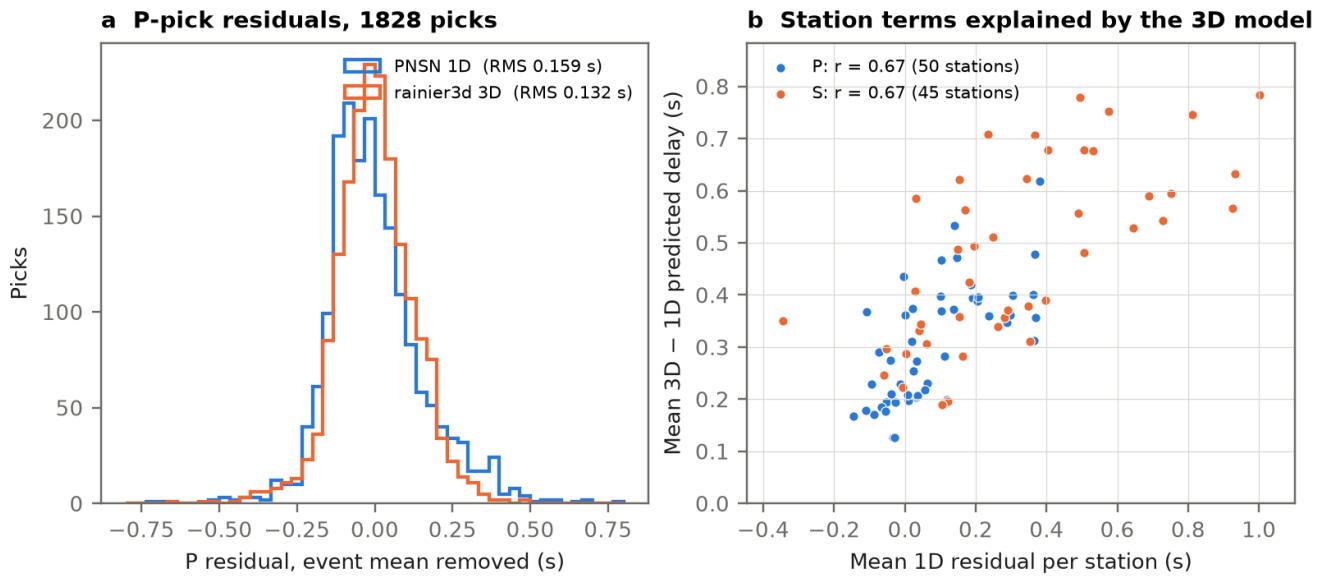


Figure 12. At the ComCat hypocentres: (a) distribution of P residuals after removing each event’s mean, for the PNSN 1D model and the fused model; (b) mean 3D – 1D predicted delay against mean 1D residual for each station with at least five picks.

- **Puget Group.** The Puget Group block west of the summit is slow down to 4 km below sea level.
 - **Seismicity.** Summit earthquakes form a column from the edifice to about 3 km below sea level. WRSZ earthquakes concentrate 4–12 km below sea level, 12–18 km west of the summit.
 - **Magma body.** The slow body at 7–10 km below sea level is largely removed by the fusion, because neither regional
- 320 model holds a slow body there at the wavelengths they resolve. Whether a body of the size imaged by Moran et al. (1999) and Pang et al. (2025) belongs in the model is a question for data that resolve it.

6 Geodetic strain and stress at depth

6.1 GNSS data and velocities

Daily GNSS positions come from two archives:

- 325 – **PANGA.** The Pacific Northwest Geodetic Array (Central Washington University) is the regional analysis centre (Pacific Northwest Geodetic Array, 2026): 188 sites in a network box of 124.0–119.8° W, 45.6–48.2° N, in its North-America-fixed frame.
- **UNR.** The Nevada Geodetic Laboratory’s IGS20 North-America-fixed series fill in 54 sites that PANGA lacks (Blewitt et al., 2018).

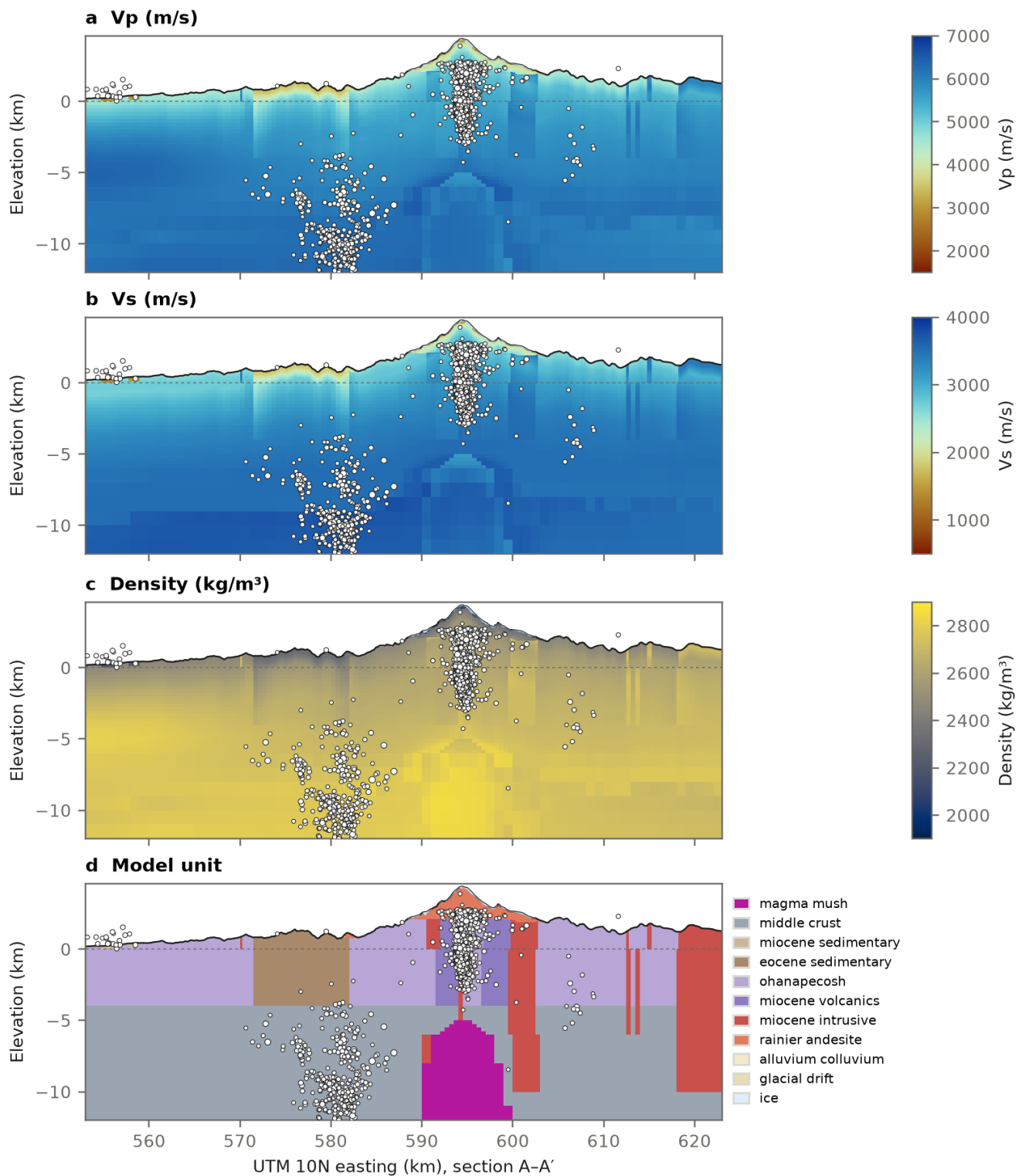


Figure 13. Fused model along A–A' (west–east through the summit): (a) Vp, (b) Vs, (c) density, (d) model units. Light blue at the surface is glacier ice. White dots are PNSN earthquakes within 2 km of the section. The dashed line is sea level.

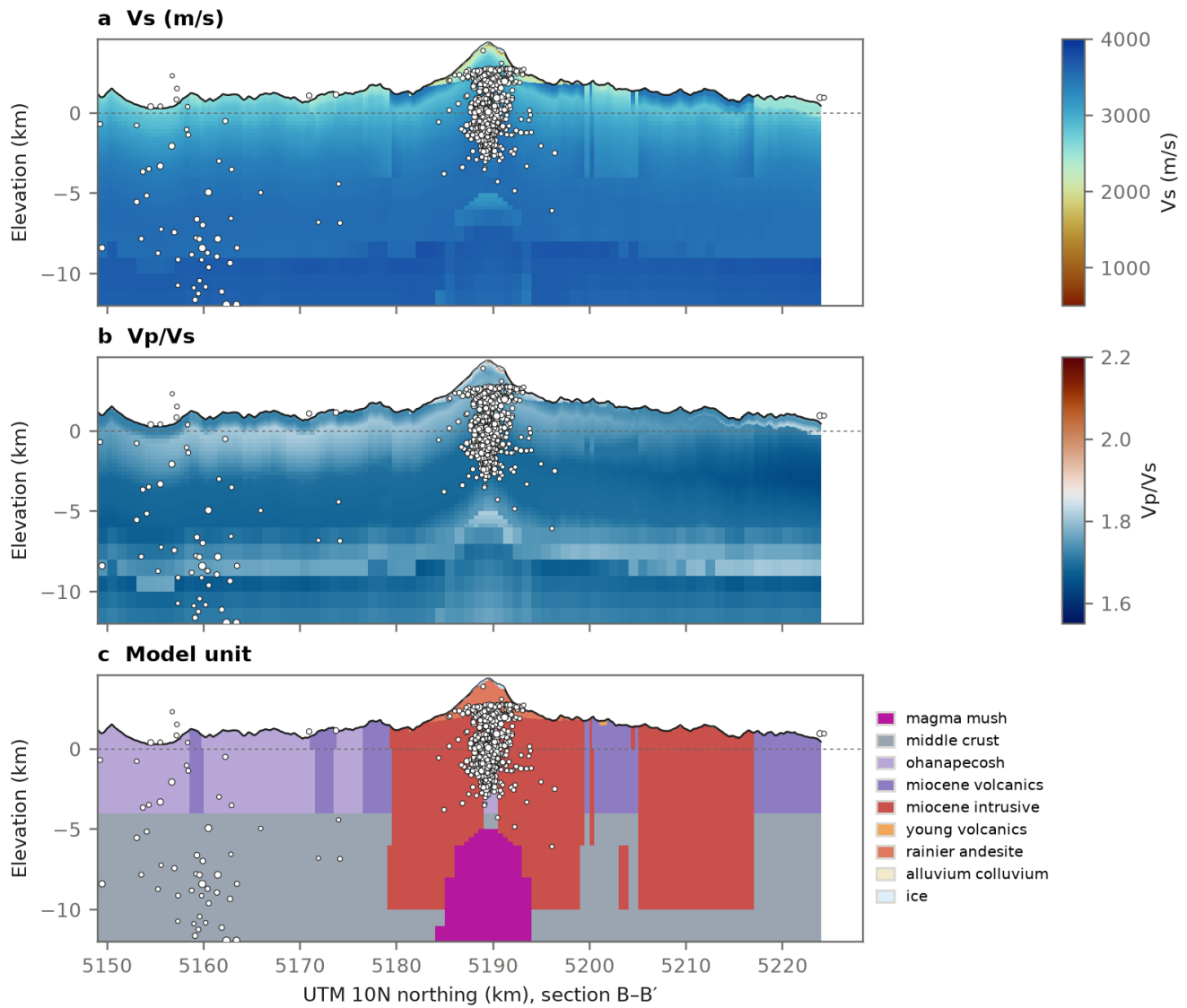


Figure 14. Fused model along B-B' (south-north through the summit): (a) Vs, (b) Vp/Vs, (c) model units. The step in Vp/Vs near 9 km below the ground marks the change from the Cascadia model to CRESCENT with Brocher's Vp.

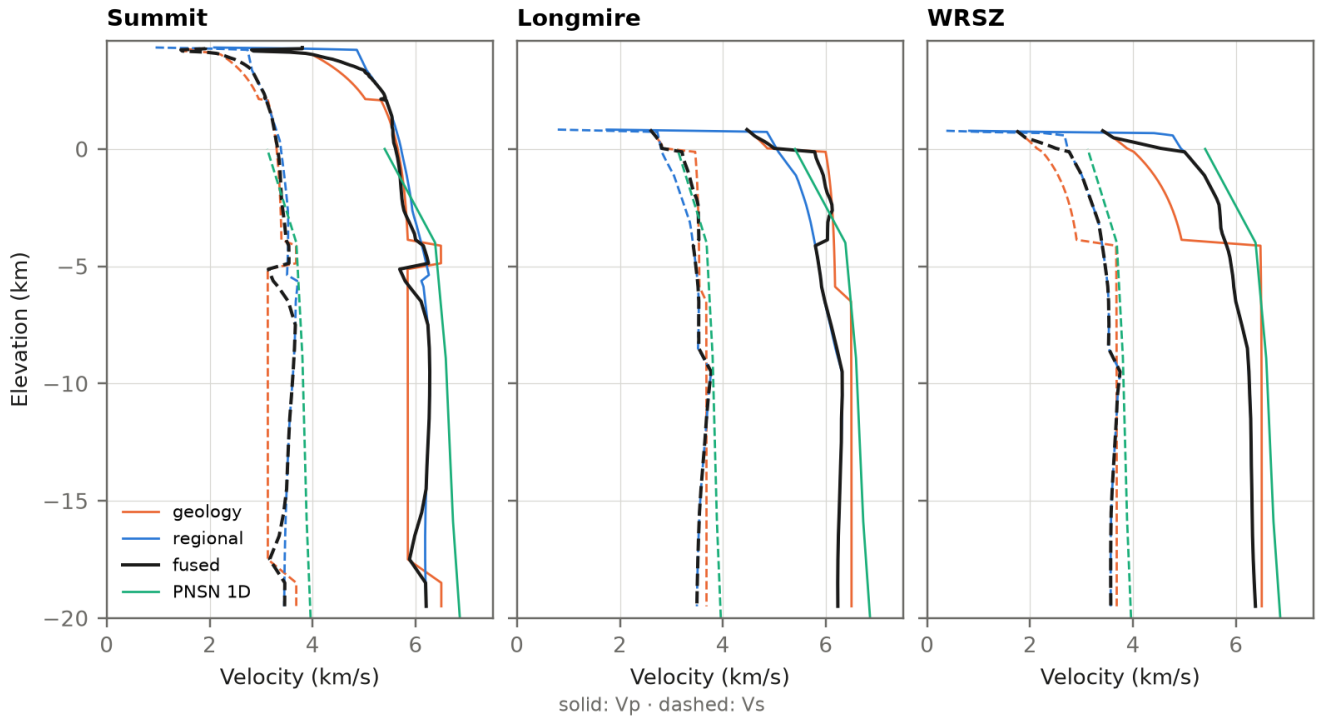


Figure 15. Vertical profiles of Vp (solid) and Vs (dashed) at the summit, at Longmire and in the WRSZ, for the geology, regional and fused models and the PNSN 1D model.

330 Together they give 1,050,643 site-days in 256 series.

- **Frame alignment.** Twenty sites that both archives process align UNR to PANGA by a translation and rotation fit, with a residual RMS of 0.10 mm yr^{-1} .
- **Velocities.** Station velocities come from MIDAS (Blewitt et al., 2016). Equipment steps are taken from the PANGA fit headers and the UNR steps database.
- 335 – **Quality control.** Sites are flagged when their velocity differs by more than 1 mm yr^{-1} from PANGA’s published velocity, or by more than 2.5 mm yr^{-1} from the median of neighbours within 40 km. This flags 22 of 203 sites. They include the high edifice sites MUIR, CSHR, SNRS, PNHG and PNHR; PNHG moves 44 mm yr^{-1} east, which is not tectonic. Flagged sites are kept in the product with their flag and left out of the strain fits.

The data are fetched again every week, and the products are recomputed (Sect. 9). The numbers below are for the series
 340 ending on 1 September 2026, which are frozen in `configs/gnss.yaml`.

6.2 Strain rate

The horizontal strain-rate tensor $\dot{\epsilon}_{ij}$ is estimated from the velocities.

- **Grid.** On a 5 km grid, it comes from a velocity-gradient fit with adaptive Gaussian weights (Shen et al., 2015). The smoothing scale grows until the summed weight and the number of stations reach their minimums.
- **Regions.** For the two regions of interest it is a uniform-strain fit to the stations around them.

The reported components are the dilatation $\dot{\epsilon}_{xx} + \dot{\epsilon}_{yy}$, the maximum shear $\sqrt{(\dot{\epsilon}_{xx} - \dot{\epsilon}_{yy})^2/4 + \dot{\epsilon}_{xy}^2}$, the principal rates and their azimuth, the rotation and the second invariant. The grid also carries the amplitude and peak day of the annual areal strain, the seasonal signal of hydrological loading.

Table 15: Secular strain rates from uniform-strain fits to MIDAS velocities (S18, `outputs/gnss/summary.json`).

Region	Sites	Dilatation (nanostrain yr ⁻¹)	Maximum shear (nanostrain yr ⁻¹)	Azimuth of $\dot{\epsilon}_1$
WRSZ (polygon + 25 km)	17	-16.6 ± 3.5	11.1	136°
Edifice (within 25 km of the summit, quality-controlled)	6	-27.9 ± 12.4	4.3	171°
Edifice, all sites	11	$+120.5 \pm 11.8$	61.2	103°

Both regions contract at a few tens of nanostrain per year (Table 15, Fig. 17a), as the rest of the model box does (median -19 nanostrain yr⁻¹). The edifice result depends on quality control: with the flagged summit sites included, it changes sign. On the grid, the summit cell has a dilatation rate of -14.6 nanostrain yr⁻¹ at a smoothing scale of 30 km, because the flagged summit sites are excluded. The annual areal-strain amplitude has a median of 15 nanostrain over the box (5–95%: 4–39), peaking in late August.

6.3 Daily strain and the summit swarms

The daily uniform strain of each region’s stations is fitted to trajectory residuals, with trend, annual and semiannual terms and steps removed per site. Stations joining or leaving the network therefore do not shift the series (Fig. 16). Across the two summit swarms, the 60-day changes are small (Table 16).

Table 16: Median absolute deviation (MAD) of the daily strain and 60-day change in areal strain across the summit swarms, in nanostrain (S18).

Region	Daily scatter (MAD)	13 February 2009 swarm	8 July 2025 swarm
WRSZ	20	+6	+7

Region	Daily scatter (MAD)	13 February 2009 swarm	8 July 2025 swarm
Edifice	106	no quality-controlled coverage	–166, inside the 411-nanostrain scatter of the 30-day medians

Neither swarm shows a transient above about 10 nanostrain in the WRSZ. On the edifice, winter excursions of up to 1500 nanostrain come from the close-spaced high sites. With stations a few kilometres apart, a few millimetres of snow-related antenna error become about 1000 nanostrain of apparent strain. Resolving strain of the edifice or of a magma body needs other instruments: tiltmeters, borehole strainmeters or interferometric synthetic-aperture radar.

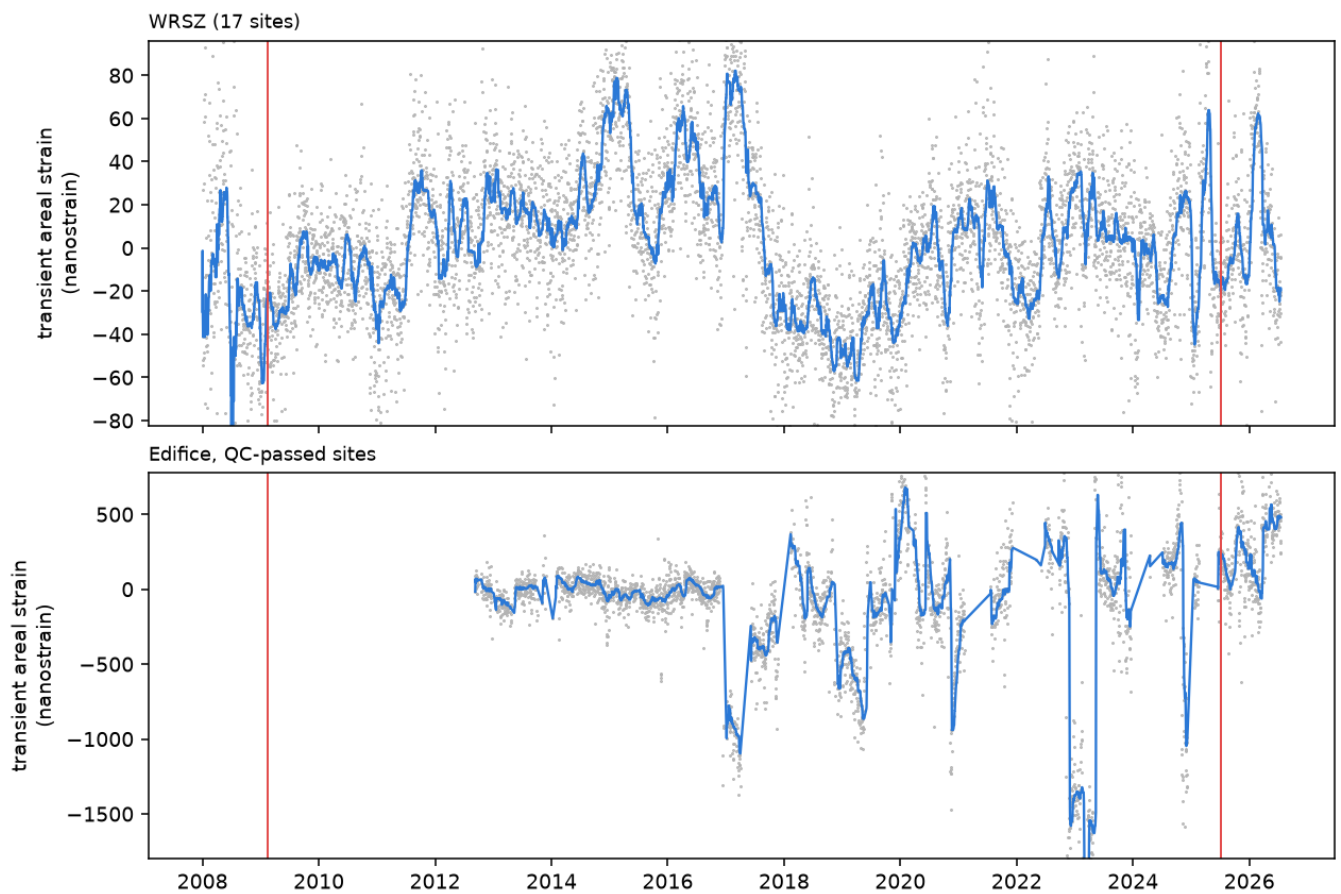


Figure 16. Daily areal strain of the stations around the WRSZ and the edifice, from trajectory residuals (grey), with 30-day medians (blue). Red lines mark the 2009 and 2025 summit swarms.

6.4 Stress from the edifice load

The weight of the edifice is a static load on the crust beneath it.

- **Load.** The edifice is taken as everything above the pre-volcanic surface of the geology model. It is represented by Boussinesq point loads on an elastic half-space (Johnson, 1985): 1131 cells of 500 m, with a total weight of 2.34×10^{15} N (95 km^3 of rock at 2500 kg m^{-3}) on a reference plane at 1789 m.
- **Output.** The six stress components, the mean stress, the maximum shear and the principal stresses are computed on the model levels L1–L3.
- **Checks.** The vertical force on a buried plane balances the load within 1%, and the stress field is in equilibrium away from the axis.

Table 17: Stress from the edifice load beneath the summit (compression negative; S18, `edifice_load.zarr`).

Elevation (m)	Vertical stress (MPa)	Mean stress (MPa)	Maximum shear (MPa)
0	−38.5	−18.7	15.2
−2000	−24.0	−9.4	11.2
−5000	−13.0	−4.4	6.6
−11,500 (magma-body centre)	−4.9	−1.5	2.6

The load stress decays from tens of megapascals beneath the summit to a few megapascals at the depth of the magma body (Table 17, Fig. 17b). That is three to four orders of magnitude above the tectonic stress rate implied by the geodetic strain rates (about 1 kPa yr^{-1} for a shear modulus of 30 GPa and $3 \times 10^{-8} \text{ yr}^{-1}$). The load stress therefore sets the orientation of stresses in the upper crust beneath the edifice. The model is a homogeneous half-space with a flat reference plane, and its Poisson’s ratio (0.25) and density (2500 kg m^{-3}) are author choices.

7 Geohydrology

Water controls much of what `rainier3d` describes:

- the seismic velocities of the shallow crust;
- the conductivity that maps alteration;
- the seasonal loading seen by GNSS;
- the lahars and debris flows of Sect. 8.

The model compiles the hydrological information needed for a hydrological description of the mountain. It holds no hydrological state: it neither models groundwater flow nor assimilates hydrological observations. Table 18 lists what it holds.

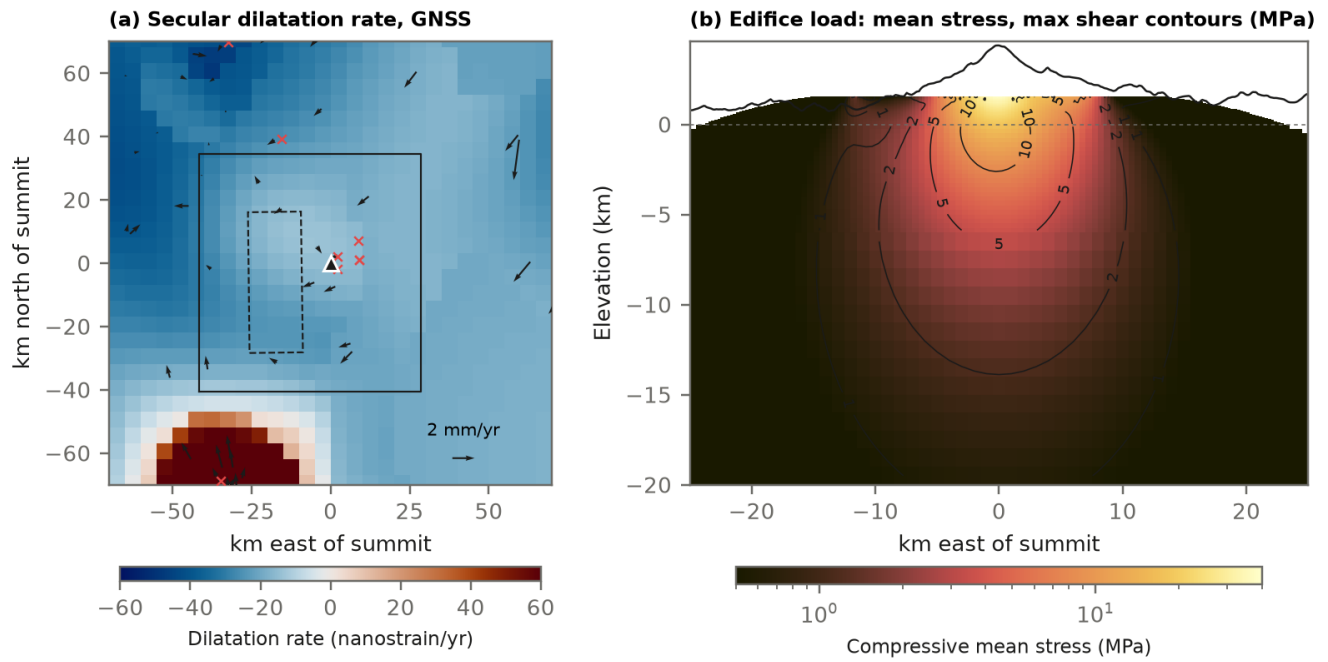


Figure 17. (a) Secular dilatation rate from the GNSS velocities, with the velocities of quality-controlled sites relative to the network median (arrows) and flagged sites (red crosses). The solid box is the model domain and the dashed polygon the WRSZ region. (b) Compressive mean stress from the edifice load along a west–east section through the summit, with contours of maximum shear (MPa).

Table 18: Hydrological information in rainier3d.

Component	Layer in rainier3d	Source	Section
Ice	glacier thickness and bed, 5.50 km ³ of ice	IceBoost v2, RGI 6.0	Sect. 4.1
Snow	NDSI at the end of the melt season	Sentinel-2	Sect. 4.2
Surface water	60,174 stream flowlines with Strahler order and mean annual flow	NHDPlus HR	Sect. 4.2
Soil storage	soil thickness (≥ 2 m capped)	SOLUS100	Sect. 4.2
Groundwater	water-table depth, two estimates	Ma et al. (2026); Fan et al. (2017)	Sect. 4.2
Interception and transpiration	canopy height, cover, leaf and plant area, biomass	lidar, Sentinel-2, GEDI	Sect. 4.3
Hydrothermal fluids	alteration intensity of the top ~150 m	1996 helicopter EM survey	Sect. 5.2
Hydrological loading	annual areal-strain amplitude and phase	GNSS	Sect. 6

The water table. The two estimates differ in level and in pattern.

- 385 – **Ma et al. (2026).** The machine-learning estimate of Ma et al. (2026b) keeps the water table within about 20 m of the ground almost everywhere in the box (median 10 m).
- **Fan et al. (2017).** The global groundwater model of Fan et al. (2017) places it hundreds of metres down under ridges.

For the shallow seismic model the choice matters more than either value. A water table separates dry from saturated rock, and saturation raises V_p and V_p/V_s below it. The Ma et al. grid is used under its CC-BY-NC-ND licence for non-commercial
390 research and is not redistributed. Its ensemble uncertainty is served through the HydroGEN platform and needs an account.

Groundwater observations. No aquifer maps exist for the park. The groundwater study of the upper White River (Fuhrig et al., 2024) is the only park-scale assessment we found. The surficial hydrothermal system described by Frank (1995) is represented only through the alteration it left.

How the layers can enter the seismic model.

- 395 – **Fluid substitution.** A water table would separate dry from saturated cells in Gassmann-type fluid substitution.
- **Valley fill.** Valley fill along the mapped streams would appear as slow, high- V_p/V_s bodies.
- **Hydrothermal core.** A fluid-saturated core would join the alteration field.
- **Loading.** The annual strain signal and the snow and water storage could be compared as a loading model.

None of these couplings is applied in the model.

400 8 **Mass movements: landslides, lahars, debris flows and avalanches**

Mass movements are Rainier’s most frequent hazard. Debris flows and outburst floods from its glaciers recur on a scale of years (Walder and Driedger, 1995; Legg et al., 2014), rock and ice avalanches fall from its steep upper slopes (Crandell and Fahnestock, 1965), and its lahars have reached the Puget Lowland (Crandell, 1971; Vallance and Scott, 1997). This section
405 adds a time-dependent layer to the model and provides labelled sources for the seismic detection of surface events.

Scope. Four classes of event:

- landslides, including rock avalanches and edifice collapses;
- lahars;
- debris flows, including glacial outburst floods that bulk into debris flows;
- 410 – snow and ice avalanches.

For each event, the catalogue records the following:

- class and date, or age for prehistoric deposits;
- source area and runout on the model grid;

- volume where published;
- the triggering conditions where known;
- the source and its licence.

Sources that are machine-readable. Four groups can seed the catalogue directly.

- **Landslide polygons.** The Washington State Landslide Inventory Database (Washington Geological Survey, 2026b) holds lidar-protocol mapping and a compilation layer. The compilation layer includes the park-wide mapping of 448 mass movements over 37 km² by Riedel and Dorsch (2016). The U.S. Landslide Inventory (Belair et al., 2025; Mirus et al., 2020) compiles the same state data nationally.
- **Lahar hazard zones.** The lahar inundation zones of Hoblitt et al. (1998) are distributed as digital data (Schilling et al., 2008). They are hazard zones, not events.
- **Seismogenic mass movements.** The Exotic Seismic Events Catalog (Collins et al., 2025) and the western United States compilation of Allstadt et al. (2017) (CC0) list seismically recorded landslides, debris flows and outburst floods with source parameters.
- **Seismic event types.** PNSN classifies surface events in its own catalogue. These classes are not in ComCat: within 25 km of the summit, ComCat lists 14,288 events and none of them is a surface event. The curated PNSN dataset of Ni et al. (2023) and the classifiers of Kharita et al. (2026) give labelled waveforms for training detectors.

Sources in the literature. Other sources are reports and papers, from which events are digitised.

- **Lahars.** Holocene lahars and their deposits: Crandell (1971), Scott et al. (1995), and the Osceola Mudflow (Vallance and Scott, 1997).
- **Rock avalanches.** The 1963 rock avalanches from Little Tahoma Peak (Crandell and Fahnestock, 1965).
- **Debris flows and outburst floods.** Along Tahoma Creek (Walder and Driedger, 1994, 1995), and debris-flow initiation in proglacial gullies (Legg et al., 2014).
- **River response.** River aggradation measured by repeat lidar (Anderson and Pitlick, 2014).
- **Seismic signatures.** Seismic and acoustic signatures of mass movements at volcanoes (Allstadt et al., 2018), and the lahar-detection network of the Cascades Volcano Observatory (Kramer et al., 2024).
- **Snow avalanches.** Observations are held by the Northwest Avalanche Center (Northwest Avalanche Center, 2026), with programmatic access on request.

Table 19: Sources for the mass-movement catalogue.

Class	Main sources	Machine-readable
Landslides, rock avalanches	WGS inventory (with NPS mapping); U.S. Landslide Inventory; ESEC	yes (polygons; seismic events)
Lahars	Crandell (1971); Scott et al. (1995); Vallance and Scott (1997); USGS hazard zones	zones only; events from the literature
Debris flows, outburst floods	Walder and Driedger (1994, 1995); Legg et al. (2014); ESEC; PNSN surface events	partly
Snow and ice avalanches	Northwest Avalanche Center; PNSN surface events	on request

Each record carries its source and licence, like every other layer. Records from the landslide inventories and the seismic catalogues can be fetched by script. Records from the literature are entered from their tables with a citation to the page.

9 Accessing the model

9.1 Products and the client

445 The derived products are published as release assets of the code repository, under CC-BY 4.0, and listed with their URLs and SHA-256 checksums in the package catalogue `src/rainier3d/products.json`.

Table 20: Downloadable products.

Product	Content	Refresh
<code>model</code>	<code>model.zarr</code> : surface node and levels L1–L3 with Vp, Vs, density, Qp, Qs, units, alteration, and the geology and regional inputs; variables from sources that forbid redistribution are removed	with each model release
<code>gnss</code>	station velocities with quality flags, strain-rate grid, daily regional strain series, the download manifest and the cut-off date	weekly (rolling release <code>gnss-latest</code> , with dated copies)
<code>edifice_load</code>	stress from the edifice load on L1–L3	with each model release

A client is installed with the package. It is available as the `rainier3d` command and as the Python module `rainier3d.api`. It downloads a product once, checks its checksum, caches it, and writes grids for other codes:

```
pip install "git+https://github.com/Denolle-Lab/mt-rainier-digital-model"
rainier3d list                                # products, versions and licences
```

```
rainier3d export model --format specfem --bbox -122.0 46.7 -121.6 47.0 \
    --dx 250 --dz 250 --zmin -10000 --out out/tomography_model.xyz
rainier3d export model --format nll --dx 500 --dz 500 --out out/nll/rainier3d
rainier3d export surface --layers elevation soil_thickness --out out/surface/ # GeoTIFFs
rainier3d sample --lon -121.76 --lat 46.85 --depth 5000 # values at one point
rainier3d fetch gnss # latest weekly strain product

import rainier3d.api as r3
tree = r3.open_model() # downloads once, then reads the cache
g = r3.grid(tree, bbox=(-122.0, 46.7, -121.6, 47.0), dx=250, dz=250, zmin=-10000)
r3.export(g, "netcdf", "out/rainier3d_250m.nc") # also nll, emc, specfem, csv
vs = tree["L2"].to_dataset().vs.interp(x=594500, y=5189500, z=-2000) # m/s, 2 km below sea level
```

Resampled grids are interpolated linearly within each level. Cells above the ground take the value of the first rock cell beneath them, so receivers at the surface sit in rock, and a variable `air` flags them. Table 21 lists the formats.

Table 21: Export formats of `rainier3d export`.

Format	Coordinates	Use
CF netCDF4	UTM 10N x, y (m); z = elevation (m, up)	eikonal solvers (pykonal, fteikpy), custom codes
NonLinLoc 3D grids (P, S), slowness $\times$ cell size	UTM 10N km, depth down below sea level, TRANSFORM NONE	Grid2Time, NLLoc
EMC netCDF3	longitude, latitude, depth below sea level (km)	EarthScope Earth Model Collaboration tools
SPECFEM3D tomography_model.xyz	UTM 10N m, elevation up, x fastest	SPECFEM3D Cartesian
CSV	x, y, z and variables, one row per node	spreadsheets, GIS
GeoTIFF (surface layers)	UTM 10N, 100 m	GIS

9.2 Locating earthquakes in the three-dimensional model

Of the 15,660 PNSN earthquakes since 1980 in the box, 500 have ComCat hypocentres at or above the ground surface, mostly shallow edifice events. Those events were located in one-dimensional models that know neither the topography nor the slow edifice. Locating them in the three-dimensional model, with the search restricted to rock, removes the problem at its source.

With NonLinLoc (Lomax et al., 2000):

- **Grid spacing.** Use equal horizontal and vertical spacing: 250 m near the edifice or 500 m for the WRSZ.
- **Keep hypocentres in rock.** Mask the search volume below a topography grid with `LOCTOPO_SURFACE`, and leave the exported velocities unchanged; air cells carry the rock velocity below them, so travel times are unbiased. Slow air

above a rock skin is an alternative, but the skin must be at least two cells thick. With a one-cell 500 m skin, Grid2Time's finite-difference start box reaches the slow air around summit stations and delays P times at station OBSR by 0.41 s on average.

- **Stations.** Give stations in the same UTM kilometre frame, with depth = $-\text{elevation}/1000$ (GTSRCE <sta> XYZ <x_km> <y_km> <-elev_km> 0.0).
- **Travel times and location.** Compute times with GTMODE GRID3D ANGLES_NO, and locate with the equal-differential-time likelihood and oct-tree search. Model errors should grow with travel time (LOCGAU2 0.02 0.05 0.5).
- **Depths.** NonLinLoc reports depths below sea level. Depth below the ground follows from surface_elevation in the netCDF export.

9.3 The three-dimensional viewer

The viewer runs in a web browser, including on phones. It is a React and three.js application (MIT licence) published at <https://denolle-lab.github.io/mt-rainier-digital-model/>, and it draws the following on the terrain:

- the surface layers of Sect. 4;
- the alteration field and the apparent magnetisation;
- the seismic, geodetic, infrasound, tiltmeter and fibre sensors, with permanent and temporary networks separated;
- the PNSN seismicity.

Below the ground it shows Vs, Vp, Vp/Vs, density, units and alteration, on a vertical section along the terrain cut and on a horizontal depth slice. Its map data are built by scripts S8 and S11 and published as a release asset named in web/viewer/DATA_RELEASE.

10 Limitations

- **Rock-physics parameters.** The unit parameters of Table 8 are chosen within published ranges and scaled by three global multipliers, of which only V_0 is resolved by the travel times. The alteration factors and the Q rule are likewise author choices. The planned replacement is a table of laboratory and field measurements on Cascade and analogue volcanic rocks (Watters et al., 2000; Heap and Violay, 2021), against which the calibrated V_0 can be tested. Per-lithology multipliers need more stations on plutonic and sedimentary units.
- **Geometry by rules.** Contacts are vertical and unit bases are flat. Structural modelling from the map contacts and orientation measurements would replace the column rules.
- **Hidden alteration and deep bodies.**
 - The alteration field covers only the top ~150 m. The buried alteration of the upper west flank (Finn et al., 2001) is not represented.

- 490 – The magma body is a prescribed ellipsoid that the fusion largely removes.
- The Southern Washington Cascades Conductor (Stanley et al., 1996) is not represented.
- **Regional correction.** It is one depth profile for the whole box. The iMUSH comparison shows that it holds beneath Rainier but makes Vs 5–7% too fast in the south, so it should vary laterally.
- **Top 300 m.** The calibrated rock is 21% faster in Vp and 30% faster in Vs than the shallowest layer of the Cascadia model,
 495 and the travel times constrain it only beneath the stations. Near-surface Vs from dense nodal and fibre arrays, or station terms estimated with a prior on their size, can settle which is right. Station terms are not estimated in the calibration.
- **Depth coverage of the calibration.** Only 30 of the 88 events are deeper than 11 km. Larger pick sets, such as the curated PNSN dataset (Ni et al., 2023), and Rainier-specific models (Obrebski et al., 2015; Flinders and Shen, 2017) would constrain the deep correction.
- 500 – **Regional model below 9.9 km.** The regional model there is CRESCENT Vs with Brocher’s Vp; the deep level of the Cascadia model (10.8–59.4 km) is not used.
- **Resolution.** L1 is 250 m × 50 m, so thin deposits fall below the cell size.
- **Glaciers.** IceBoost exceeds the 1981 radar thicknesses on Emmons and Winthrop glaciers; its total should be compared with the lidar-based ice volume of Sisson et al. (2011).
- 505 – **Geodesy.** The GNSS network does not resolve strain on the edifice. The edifice-load model is a homogeneous half-space.
- **Hydrology.** The model has no hydrological state (Sect. 7).

11 Conclusions

rainier3d assembles, in one reproducible structure, what is openly known about the subsurface and surface of Mount Rainier.

- **Subsurface.** A geology-driven velocity model, fused with the regional tomography, calibrated on PNSN P and S–P
 510 travel times with events relocated in 3D, and validated on held-out events and an independent tomography. The calibration shows that the table rock is too slow near the surface: V_0 is raised by 31%. It also shows that the regional Vs beneath Rainier is 5–9% too slow at 2–7 km, a correction the iMUSH tomography confirms locally.
 - **Surface.** Layers for geology, ice, soil, water, vegetation and imagery on a common grid.
 - **Geodesy and load.** GNSS strain rates, refreshed weekly, and the stress of the edifice load at depth.
- 515 Every input is fetched from its original archive and checksummed. Every parameter names its source, and every product can be downloaded with one command in the formats that seismological codes read. The next layers are the mass-movement catalogue and couplings between the hydrological layers and the seismic properties.

12 Code and data availability

The code, configuration, source registry and this paper are at <https://github.com/Denolle-Lab/mt-rainier-digital-model> (BSD-3-Clause). The derived products are release assets of the same repository (CC-BY 4.0), listed with checksums in `src/rainier3d/products.json` and downloaded with `rainier3d fetch`.

Appendix A lists the input data sets with their DOIs or service addresses. `docs/data_policy.md` gives the licence tier of each. `docs/data_manifest.csv` gives the checksum of each cached input.

Two inputs require attribution notices:

- GPS time series are provided by the Pacific Northwest Geodetic Array, Central Washington University.
- The imagery layers contain modified Copernicus Sentinel data (2023, 2025).

The code and this paper were written with an AI coding assistant (Claude, Anthropic) under the direction of the authors. All numbers are produced by the scripts named in the text, and all DOIs were resolved against the Crossref and DataCite registries.

Appendix A: Data sets

Table A1: Input data sets.

Data set	Role in the model	Reference
Cascadia velocity model v1.7	regional Vp and Vs to 9.9 km	Wirth et al. (2025); Wirth Moriarty et al. (2025)
CRESCENT Gen0 community velocity model	regional Vs below 9.9 km	He et al. (2026); He (2026)
Washington 1:100,000 surface geology (GeMS)	surface units	Washington Geological Survey (2026a)
Geology of Mount Rainier National Park	stratigraphy; map overlay	Fiske et al. (1963)
USGS 3D Elevation Program	elevation	U.S. Geological Survey (2026c)
IceBoost v2 per-glacier thickness; RGI 6.0	ice thickness; glacier identifiers	Maffezzoli (2025); RGI Consortium (2017)
GlaThiDa; 1981 radar surveys	glacier thickness check	World Glacier Monitoring Service (WGMS) (2020); Driedger and Kennard (1986)
1996 helicopter EM and magnetic survey	alteration	Rystrom et al. (2000)
iMUSH local-earthquake tomography	independent check	Incorporated Research Institutions for Seismology (2020); Ulberg et al. (2020)

Data set	Role in the model	Reference
PNSN origins and phase picks (ComCat)	calibration and validation	U.S. Geological Survey (2026a)
FDSN station metadata	station coordinates, sensor inventory	EarthScope Consortium (2026)
PANGA and UNR GNSS daily positions	velocities, strain	Pacific Northwest Geodetic Array (2026); Blewitt et al. (2018)
SOLUS100	soil thickness	Nauman (2024)
US water-table depth	water table	Ma et al. (2026a)
Global water-table depth	water table, comparison	Fan et al. (2017)
NHDPlus High Resolution	streams	U.S. Geological Survey (2026b)
ETH global canopy height 2020	canopy height	Lang, Nico et al. (2022)
NLCD 2021	land cover	U.S. Geological Survey (2023)
Copernicus Sentinel-2 L2A	imagery, NDVI, NDSI, leaf area index	European Space Agency (2025); European Space Agency (2023)
GED1 L2B, L3, L4B	plant area index, canopy height, biomass	Dubayah et al. (2021b); Dubayah et al. (2021a); Dubayah et al. (2025)

530 *Author contributions.* MD designed the model, the calibration and the validation and directed the work. DY designed and built the three-dimensional viewer.

Competing interests. The authors declare that they have no conflict of interest.

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